# Projections and Linear Systems

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## Design Matrix

**Definition 0.1** (Design matrix (model matrix))**.** In a regression model with $n$ observations and $p$ predictors, the **design matrix** [1] (or **model matrix**) $\mathbf{X}$ is the $n \times p$ matrix whose $i$-th row is the covariate vector $\underset{\sim}{x}_i^\top$ for observation $i$:

$$\mathbf{X} = \begin{bmatrix} \underset{\sim}{x}_1^\top \\ \underset{\sim}{x}_2^\top \\ \vdots \\ \underset{\sim}{x}_n^\top \end{bmatrix} = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix}$$

Remark (Products with the design matrix)

*Remark* 0.1 (Products with the design matrix). The product $\mathbf{X}\underset{\sim}{\beta}$ collects the values $\underset{\sim}{x}_i^\top \underset{\sim}{\beta}$ for all $n$ observations into a single $n \times 1$ vector:

$$\mathbf{X}\underset{\sim}{\beta} = \begin{bmatrix} \underset{\sim}{x}_1^\top \underset{\sim}{\beta} \\ \vdots \\ \underset{\sim}{x}_n^\top \underset{\sim}{\beta} \end{bmatrix}$$

For example, take the rank-2 matrix from Example 2 in Subspaces and Rank as $\mathbf{X}$, with $n = 3$ observations and $p = 2$ columns: an intercept column and one covariate. With $\underset{\sim}{\beta} = (1, 2)$,

$$\mathbf{X}\underset{\sim}{\beta} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$
$$= \begin{bmatrix} 3 \\ 5 \\ 7 \end{bmatrix}.$$

The matrix $\mathbf{X}^\top\mathbf{X}$ is $p \times p$ and symmetric, since

$$\begin{aligned}(\mathbf{X}^\top\mathbf{X})^\top &= \mathbf{X}^\top(\mathbf{X}^\top)^\top \\ &= \mathbf{X}^\top\mathbf{X}\end{aligned}$$

(Theorem 16 in Matrices). When $\mathbf{X}^\top\mathbf{X}$ is invertible, it appears in the least squares solution (Definition 0.10, defined later in this chapter) $\underset{\sim}{\hat{\beta}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\underset{\sim}{y}$.

**Theorem 0.1** ($\mathbf{X}^\top\mathbf{X}$ is invertible when $\mathbf{X}$ has full column rank)**.** *If $\mathbf{X}$ is an $n \times p$ matrix with $\operatorname{rank}(\mathbf{X}) = p$ (Definition 2 in Subspaces and Rank), then the $p \times p$ matrix $\mathbf{X}^\top\mathbf{X}$ is invertible.*

Proof

*Proof.* Let $\underset{\sim}{c}$ be a vector of length $p$ with $\mathbf{X}^\top\mathbf{X}\underset{\sim}{c} = \underset{\sim}{0}$. Then

$$\begin{aligned} 0 &= \underset{\sim}{c}^\top\mathbf{X}^\top\mathbf{X}\underset{\sim}{c} && \text{(multiply } \mathbf{X}^\top\mathbf{X}\underset{\sim}{c} = \underset{\sim}{0} \text{ on the left by } \underset{\sim}{c}^\top\text{)} \\ &= (\mathbf{X}\underset{\sim}{c})^\top(\mathbf{X}\underset{\sim}{c}) && \text{(transpose of a product)} \\ &= \|\mathbf{X}\underset{\sim}{c}\|^2 && \text{(definition of the Euclidean norm)} \end{aligned}$$

so $\mathbf{X}\underset{\sim}{c} = \underset{\sim}{0}$. Because $\mathbf{X}\underset{\sim}{c} = c_1(\text{column } 1) + \cdots + c_p(\text{column } p)$ and the columns of $\mathbf{X}$ are linearly independent, $\underset{\sim}{c} = \underset{\sim}{0}$. So the only solution of $\mathbf{X}^\top\mathbf{X}\underset{\sim}{c} = \underset{\sim}{0}$ is $\underset{\sim}{c} = \underset{\sim}{0}$, which says that the columns of the square matrix $\mathbf{X}^\top\mathbf{X}$ are linearly independent, and a square matrix with linearly independent columns is invertible [2], Corollary 5.7, p. 143. □

**Example 0.1** (Inverting $\mathbf{X}^\top\mathbf{X}$)**.** For the rank-2 matrix $\mathbf{X} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$ from Example 2 in Subspaces and Rank:

$$\mathbf{X}^\top\mathbf{X} = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}, \qquad (\mathbf{X}^\top\mathbf{X})^{-1} = \frac{1}{6}\begin{bmatrix} 14 & -6 \\ -6 & 3 \end{bmatrix},$$

and multiplying the two out gives $\mathbf{I}_2$.

**Definition 0.2** (Hat matrix)**.** For an $n \times p$ design matrix $\mathbf{X}$ (Definition 0.1) with $\operatorname{rank}(\mathbf{X}) = p$, the **hat matrix** [3] is the $n \times n$ matrix

$$\underbrace{\mathbf{H}}_{n\times n} \overset{\text{def}}{=} \underbrace{\mathbf{X}}_{n\times p} \underbrace{(\mathbf{X}^\top \mathbf{X})^{-1}}_{p\times p} \underbrace{\mathbf{X}^\top}_{p\times n}$$

The inverse exists by Theorem 0.1.

**Example 0.2** (The hat matrix of an intercept-only model)**.** With $n = 2$ observations and only an intercept, $\mathbf{X} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ ($2 \times 1$, rank 1), so $\mathbf{X}^\top \mathbf{X} = 2$ and

$$\begin{aligned} \mathbf{H} &= \begin{bmatrix} 1 \\ 1 \end{bmatrix} \cdot \frac{1}{2} \cdot \begin{bmatrix} 1 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix}. \end{aligned}$$

Then $\mathbf{H}y = (\bar{y}, \bar{y})$, where $\bar{y} = (y_1 + y_2)/2$: the fitted values of an intercept-only model are the sample mean.

**Theorem 0.2** (Hat matrix is a projection matrix)**.** *If $\mathbf{X}$ is an $n \times p$ design matrix with $\operatorname{rank}(\mathbf{X}) = p$, then the hat matrix $\mathbf{H}$ (Definition 0.2) is an orthogonal projection matrix (Definition 8 in Special Matrices and Decompositions).*

Proof

*Proof.* We verify symmetry and idempotency. Both use that $\mathbf{X}^\top \mathbf{X}$ is symmetric,

$$\begin{aligned} (\mathbf{X}^\top \mathbf{X})^\top &= \mathbf{X}^\top (\mathbf{X}^\top)^\top \\ &= \mathbf{X}^\top \mathbf{X} \end{aligned}$$

(Theorem 16 in Matrices), so its inverse is symmetric too (Corollary 1 in Special Matrices and Decompositions).

**Symmetry:**

$$\begin{aligned} \mathbf{H}^\top &= (\mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top)^\top && \text{(definition of } \mathbf{H}) \\ &= (\mathbf{X}^\top)^\top \cdot ((\mathbf{X}^\top \mathbf{X})^{-1})^\top \cdot \mathbf{X}^\top && \text{(transpose of a product, twice)} \\ &= \mathbf{X} \cdot ((\mathbf{X}^\top \mathbf{X})^{-1})^\top \cdot \mathbf{X}^\top && \text{(transposing twice changes nothing)} \\ &= \mathbf{X} \cdot (\mathbf{X}^\top \mathbf{X})^{-1} \cdot \mathbf{X}^\top && \text{(the inverse of a symmetric matrix is symmetric)} \\ &= \mathbf{H} && \text{(definition of } \mathbf{H}) \end{aligned}$$

**Idempotency:**

$$\begin{aligned}
\mathbf{H}^2 &= \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top \cdot \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top && \text{(definition of } \mathbf{H}\text{)} \\
&= \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}(\mathbf{X}^\top\mathbf{X})(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top && \text{(regroup; matrix multiplication is associative)} \\
&= \mathbf{X}\,\mathbf{I}_p\,(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top && \text{(definition of the inverse)} \\
&= \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top && \text{(identity matrix)} \\
&= \mathbf{H} && \text{(definition of } \mathbf{H}\text{)}
\end{aligned}$$

□

**Example 0.3** (The intercept-only hat matrix is a projection)**.** For $\mathbf{H} = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix}$ from Example 0.2, $\mathbf{H}^\top = \mathbf{H}$, and

$$\begin{aligned}
\mathbf{H}^2 &= \begin{bmatrix} 0.5 \cdot 0.5 + 0.5 \cdot 0.5 & 0.5 \cdot 0.5 + 0.5 \cdot 0.5 \\ 0.5 \cdot 0.5 + 0.5 \cdot 0.5 & 0.5 \cdot 0.5 + 0.5 \cdot 0.5 \end{bmatrix} \\
&= \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix} \\
&= \mathbf{H},
\end{aligned}$$

as Theorem 0.2 says.

ℹ Remark (Why it is called the hat matrix)

*Remark* 0.2 (Why it is called the hat matrix). The hat matrix gives the fitted values of the least squares solution (Definition 0.10, defined later in this chapter):

$$\begin{aligned}
\underset{\sim}{\widehat{y}} &= \mathbf{X}\widehat{\beta} \\
&= \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top \underset{\sim}{y} \\
&= \mathbf{H}\underset{\sim}{y}.
\end{aligned}$$

Multiplying by $\mathbf{H}$ "puts a hat" on $y$, which is where the name comes from. In Example 0.2, $\mathbf{H}$ puts a hat on $\underset{\sim}{y}$ by replacing each $y_i$ with the sample mean $\bar{y}$.

### Orthogonal projection onto a subspace

> **Source**
>
> This section is adapted from the second half of Zhou [4], used under the MIT License (see the license text in Section 2 in Subspaces and Rank). It connects the orthogonal projection matrices of Definition 8 in Special Matrices and Decompositions, defined by an algebraic property, to the geometric decomposition of Theorem 8 in Direct Sums and Orthogonal Complements.

**Definition 0.3** (Orthogonal projection onto a subspace)**.** Let $\mathcal{S}$ be a subspace of $\mathbb{R}^p$ and $y \in \mathbb{R}^p$. By Theorem 8 in Direct Sums and Orthogonal Complements, $y = u + v$ for exactly one $u \in \mathcal{S}$ and $v \in \mathcal{S}^\perp$. The vector $u$ is the **orthogonal projection** of $y$ onto $\mathcal{S}$. It is the projection of $y$ onto $\mathcal{S}$ along $\mathcal{S}^\perp$ (Definition 4 in Direct Sums and Orthogonal Complements).

**Example 0.4** (Projecting onto a line in $\mathbb{R}^3$)**.** In Example 15 in Direct Sums and Orthogonal Complements, $y = (3, 1, 2)$ splits as $(2, 2, 0) + (1, -1, 2)$, with $(2, 2, 0) \in \mathcal{S} = \text{span}\{(1, 1, 0)\}$ and $(1, -1, 2) \in \mathcal{S}^\perp$. So the orthogonal projection of $(3, 1, 2)$ onto $\mathcal{S}$ is $(2, 2, 0)$. A vector already in $\mathcal{S}$, such as $(5, 5, 0)$, is its own projection, since $(5, 5, 0) = (5, 5, 0) + 0$ and $0 \in \mathcal{S}^\perp$ (Theorem 6 in Direct Sums and Orthogonal Complements, Theorem 1 in Subspaces and Rank); a vector of $\mathcal{S}^\perp$, such as $(1, -1, 2)$, projects to $0$, since $(1, -1, 2) = 0 + (1, -1, 2)$ and $0 \in \mathcal{S}$ (Theorem 1 in Subspaces and Rank).

**Example 0.5** (A split that is not the orthogonal one)**.** $(3, 1, 2) = (3, 3, 0) + (0, -2, 2)$ also writes $(3, 1, 2)$ as a vector of $\mathcal{S} = \text{span}\{(1, 1, 0)\}$ plus a remainder, but $(1, 1, 0) \cdot (0, -2, 2) = -2 \neq 0$, so the remainder is not in $\mathcal{S}^\perp$, and $(3, 3, 0)$ is not the orthogonal projection of $(3, 1, 2)$ onto $\mathcal{S}$.

**Theorem 0.3** (The orthogonal projection is the closest point)**.** *Let $\mathcal{S}$ be a subspace of $\mathbb{R}^p$, let $y \in \mathbb{R}^p$, and let $u$ be the orthogonal projection of $y$ onto $\mathcal{S}$ (Definition 0.3). Then for every $w \in \mathcal{S}$,*

$$\|y - u\| \leq \|y - w\|,$$

*with equality only when $w = u$.*

Proof

*Proof.* By Definition 0.3, $y - u \in \mathcal{S}^{\perp}$. Both $u$ and $w$ are in $\mathcal{S}$, so $u - w = u + (-1)\, w \in \mathcal{S}$ (Definition 4 in Subspaces and Rank), and therefore $(y - u) \perp (u - w)$ (Definition 5 in Direct Sums and Orthogonal Complements). Then

$$\begin{aligned} \|y - w\|^2 &= \|(y - u) + (u - w)\|^2 && \text{(add and subtract } u\text{)} \\ &= \|y - u\|^2 + \|u - w\|^2 && \text{(Pythagorean theorem, Theorem 2 in Inner Products and Orthogonality)} \\ &\geq \|y - u\|^2, && (\|u - w\|^2 \geq 0, \text{ Theorem 1 in Inner Products and Orthogonality}) \end{aligned}$$

with equality exactly when $\|u - w\| = 0$, that is, when $w = u$ (Theorem 1 in Inner Products and Orthogonality, part 1). Taking nonnegative square roots preserves the inequality. □

**Example 0.6** (No point of the line is closer)**.** In Example 0.4, $y = (3, 1, 2)$ has projection $u = (2, 2, 0)$ onto $\mathcal{S} = \text{span}\{(1, 1, 0)\}$, at distance $\|(1, -1, 2)\| = \sqrt{6}$. Another point of $\mathcal{S}$, such as $w = (3, 3, 0)$, is farther:

$$\begin{aligned} \|y - w\| &= \|(0, -2, 2)\| \\ &= \sqrt{8} \\ &> \sqrt{6}, \end{aligned}$$

and the difference of squares is

$$\begin{aligned} \|u - w\|^2 &= \|(-1, -1, 0)\|^2 \\ &= 2 \\ &= 8 - 6, \end{aligned}$$

by the Pythagorean theorem (Theorem 2 in Inner Products and Orthogonality), since $y - u \in \mathcal{S}^{\perp}$ and $u - w \in \mathcal{S}$.

**Theorem 0.4** (Projecting with an orthonormal basis)**.** *Let $\mathcal{S}$ be a subspace of $\mathbb{R}^p$ with $\dim(\mathcal{S}) = r \geq 1$, and let $\mathbf{Q}$ be the $p \times r$ matrix whose columns $q_1, \ldots, q_r$ are an orthonormal basis of $\mathcal{S}$ (Definition 10 in Inner Products and Orthogonality, Corollary 4 in Inner Products and Orthogonality). Then for every $y \in \mathbb{R}^p$, $\mathbf{Q}\mathbf{Q}^{\top} y$ is the orthogonal projection of $y$ onto $\mathcal{S}$, and $\mathbf{Q}\mathbf{Q}^{\top}$ is an orthogonal projection matrix (Definition 8 in Special Matrices and Decompositions).*

Proof

*Proof.* **$\mathbf{Q}^\top\mathbf{Q} = \mathbf{I}_r$.** Write $q_{ki}$ for entry $k$ of $\underset{\sim}{q}_i$, which is entry $(k, i)$ of $\mathbf{Q}$. Then

$$\begin{aligned}(\mathbf{Q}^\top\mathbf{Q})_{ij} &= \sum_{k=1}^{p}(\mathbf{Q}^\top)_{ik}\, q_{kj} && \text{(Definition 7 in Matrices)}\\ &= \sum_{k=1}^{p} q_{ki}\, q_{kj} && \text{(Definition 3 in Matrices)}\\ &= \underset{\sim}{q}_i \cdot \underset{\sim}{q}_j, && \text{(Definition 8 in Vectors)}\end{aligned}$$

which is 1 if $i = j$ and 0 otherwise (Definition 19 in Vectors), so $\mathbf{Q}^\top\mathbf{Q} = \mathbf{I}_r$ (Definition 12 in Matrices).

**The split.** Let $\underset{\sim}{u} = \mathbf{Q}\mathbf{Q}^\top\underset{\sim}{y}$ and $\underset{\sim}{v} = \underset{\sim}{y} - \underset{\sim}{u}$, so $\underset{\sim}{y} = \underset{\sim}{u} + \underset{\sim}{v}$. $\underset{\sim}{u} = \mathbf{Q}\,(\mathbf{Q}^\top\underset{\sim}{y})$ (Theorem 5 in Matrices) is in $\mathcal{C}(\mathbf{Q})$ (Definition 1 in Column Space, Null Space and Rank-Nullity), which is $\text{span}\{\underset{\sim}{q}_1, \ldots, \underset{\sim}{q}_r\} = \mathcal{S}$ (Theorem 1 in Column Space, Null Space and Rank-Nullity). For $\underset{\sim}{v}$,

$$\begin{aligned}\mathbf{Q}^\top\underset{\sim}{v} &= \mathbf{Q}^\top(\underset{\sim}{y} - \mathbf{Q}\mathbf{Q}^\top\underset{\sim}{y}) && \text{(substitute)}\\ &= \mathbf{Q}^\top\underset{\sim}{y} - \mathbf{Q}^\top(\mathbf{Q}\mathbf{Q}^\top\underset{\sim}{y}) && \text{(Theorem 4 in Subspaces and Rank, with coefficients 1 and } -1)\\ &= \mathbf{Q}^\top\underset{\sim}{y} - (\mathbf{Q}^\top\mathbf{Q})\,\mathbf{Q}^\top\underset{\sim}{y} && \text{(Theorem 5 in Matrices)}\\ &= \mathbf{Q}^\top\underset{\sim}{y} - \mathbf{I}_r\,\mathbf{Q}^\top\underset{\sim}{y} && (\mathbf{Q}^\top\mathbf{Q} = \mathbf{I}_r\text{, from the first part of the proof})\\ &= \mathbf{Q}^\top\underset{\sim}{y} - \mathbf{Q}^\top\underset{\sim}{y} && \text{(Theorem 7 in Matrices)}\\ &= \underset{\sim}{0}_r, && \text{(arithmetic)}\end{aligned}$$

so $\underset{\sim}{v} \in \mathcal{N}(\mathbf{Q}^\top)$, and

$$\begin{aligned}\mathcal{N}(\mathbf{Q}^\top) &= \mathcal{C}(\mathbf{Q})^\perp\\ &= \mathcal{S}^\perp\end{aligned}$$

(Theorem 7 in Direct Sums and Orthogonal Complements). By the uniqueness in Theorem 8 in Direct Sums and Orthogonal Complements, $\underset{\sim}{u}$ is the orthogonal projection of $\underset{\sim}{y}$ onto $\mathcal{S}$.

**$\mathbf{Q}\mathbf{Q}^\top$ is symmetric and idempotent.**

$$\begin{aligned}(\mathbf{Q}\mathbf{Q}^\top)^\top &= (\mathbf{Q}^\top)^\top\,\mathbf{Q}^\top && \text{(Theorem 16 in Matrices)}\\ &= \mathbf{Q}\mathbf{Q}^\top, && \text{(transposing twice, Definition 3 in Matrices)}\end{aligned}$$

and

$$\begin{aligned}(\mathbf{Q}\mathbf{Q}^\top)(\mathbf{Q}\mathbf{Q}^\top) &= \mathbf{Q}\,(\mathbf{Q}^\top\mathbf{Q})\,\mathbf{Q}^\top && \text{(Theorem 5 in Matrices)}\\ &= \mathbf{Q}\,\mathbf{I}_r\,\mathbf{Q}^\top && (\mathbf{Q}^\top\mathbf{Q} = \mathbf{I}_r\text{, from the first part of the proof})\\ &= \mathbf{Q}\mathbf{Q}^\top. && \text{(Theorem 7 in Matrices)}\end{aligned}$$

□

**Example 0.7** (The projection matrix onto a line)**.** For $\mathcal{S} = \text{span}\{(1,1,0)\}$, the single vector $\underset{\sim}{q}_1 = \frac{1}{\sqrt{2}}(1,1,0)$ is an orthonormal basis: it spans $\mathcal{S}$, it is nonzero and so linearly independent, and

$$\begin{aligned}\|\underset{\sim}{q}_1\| &= \sqrt{\tfrac{1}{2} + \tfrac{1}{2} + 0} \\ &= 1.\end{aligned}$$

With $\mathbf{Q} = [\underset{\sim}{q}_1]$,

$$\begin{aligned}\mathbf{Q}\mathbf{Q}^\top &= \tfrac{1}{2}\begin{bmatrix}1\\1\\0\end{bmatrix}\begin{bmatrix}1 & 1 & 0\end{bmatrix} \\ &= \begin{bmatrix}\frac{1}{2} & \frac{1}{2} & 0\\ \frac{1}{2} & \frac{1}{2} & 0\\ 0 & 0 & 0\end{bmatrix}.\end{aligned}$$

Applied to $\underset{\sim}{y} = (3,1,2)$, it gives $\left(\frac{3+1}{2}, \frac{3+1}{2}, 0\right) = (2,2,0)$, the projection found in Example 0.4.

**Theorem 0.5** (A projection matrix projects onto its column space, and is the only one that does)**.**

1. *If* $\mathbf{P}$ *is a* $p \times p$ *orthogonal projection matrix (Definition 8 in Special Matrices and Decompositions), then for every* $\underset{\sim}{y} \in \mathbb{R}^p$, $\mathbf{P}\underset{\sim}{y}$ *is the orthogonal projection of* $\underset{\sim}{y}$ *onto* $\mathcal{C}(\mathbf{P})$ *(Definition 0.3).*
2. *If* $\mathbf{P}_1$ *and* $\mathbf{P}_2$ *are* $p \times p$ *matrices such that, for every* $\underset{\sim}{y}$*, both* $\mathbf{P}_1\underset{\sim}{y}$ *and* $\mathbf{P}_2\underset{\sim}{y}$ *are the orthogonal projection of* $\underset{\sim}{y}$ *onto the same subspace* $\mathcal{S}$*, then* $\mathbf{P}_1 = \mathbf{P}_2$*.*

Proof

*Proof.* **Part 1.** Write $\underset{\sim}{y} = \mathbf{P}\underset{\sim}{y} + (\underset{\sim}{y} - \mathbf{P}\underset{\sim}{y})$. The first term is in $\mathcal{C}(\mathbf{P})$ (Definition 1 in Column Space, Null Space and Rank-Nullity). For the second,

$$\begin{aligned}
\mathbf{P}\,(\underset{\sim}{y} - \mathbf{P}\underset{\sim}{y}) &= \mathbf{P}\underset{\sim}{y} - \mathbf{P}\,(\mathbf{P}\underset{\sim}{y}) && \text{(Theorem 4 in Subspaces and Rank, with coefficients 1 and } -1\text{)} \\
&= \mathbf{P}\underset{\sim}{y} - (\mathbf{P}\mathbf{P})\underset{\sim}{y} && \text{(Theorem 5 in Matrices)} \\
&= \mathbf{P}\underset{\sim}{y} - \mathbf{P}^2\underset{\sim}{y} && \text{(Definition 2 in Special Matrices and Decompositions)} \\
&= \mathbf{P}\underset{\sim}{y} - \mathbf{P}\underset{\sim}{y} && (\mathbf{P} \text{ is idempotent}) \\
&= \underset{\sim}{0}_p, && \text{(arithmetic)}
\end{aligned}$$

so $\underset{\sim}{y} - \mathbf{P}\underset{\sim}{y} \in \mathcal{N}(\mathbf{P})$. Since $\mathbf{P} = \mathbf{P}^\top$,

$$\begin{aligned}
\mathcal{N}(\mathbf{P}) &= \mathcal{N}(\mathbf{P}^\top) \\
&= \mathcal{C}(\mathbf{P})^\perp
\end{aligned}$$

(Theorem 7 in Direct Sums and Orthogonal Complements). By the uniqueness in Theorem 8 in Direct Sums and Orthogonal Complements, $\mathbf{P}\underset{\sim}{y}$ is the orthogonal projection of $\underset{\sim}{y}$ onto $\mathcal{C}(\mathbf{P})$.

**Part 2.** The orthogonal projection of a vector onto $\mathcal{S}$ is unique (Definition 0.3), so $\mathbf{P}_1\underset{\sim}{e}_j = \mathbf{P}_2\underset{\sim}{e}_j$ for each indicator vector $\underset{\sim}{e}_j$ (Definition 14 in Vectors). $\mathbf{P}_1\underset{\sim}{e}_j$ is column $j$ of $\mathbf{P}_1$: by Theorem 3 in Subspaces and Rank it is the combination of the columns of $\mathbf{P}_1$ with coefficients the entries of $\underset{\sim}{e}_j$, which are 1 in place $j$ and 0 elsewhere (Definition 14 in Vectors). Likewise for $\mathbf{P}_2$, so the two matrices have the same columns. □

**Example 0.8** (Two routes to the same projection matrix)**.** The matrix $\mathbf{P} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ of Example 12 in Special Matrices and Decompositions is an orthogonal projection matrix, and its column space is span$\{(1,0)\}$, the horizontal axis. By part 1, $\mathbf{P}(v_1, v_2) = (v_1, 0)$ is the orthogonal projection onto that axis: the remainder $(0, v_2)$ is orthogonal to $(1,0)$. The vector $\underset{\sim}{q}_1 = (1,0)$ has length 1 and spans the axis, so it is an orthonormal basis of it, and with $\mathbf{B} = [\underset{\sim}{q}_1]$,

$$\begin{aligned}
\mathbf{B}\mathbf{B}^\top &= \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \end{bmatrix} \\
&= \mathbf{P},
\end{aligned}$$

as part 2 and Theorem 0.4 require.

The oblique projection $\mathbf{Q} = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$ of Example 12 in Special Matrices and Decompositions also maps every vector into the horizontal axis. It sends $(0,1)$ to $(1,0)$, but the remainder $(0,1) - (1,0) = (-1,1)$ is not orthogonal to the axis, so $(1,0)$ is not the orthogonal projection of $(0,1)$ onto the axis, which is $\mathbf{P}(0,1) = (0,0)$.

**Theorem 0.6** (The hat matrix projects onto the column space of the design matrix)**.** *If* $\mathbf{X}$ *is an* $n \times p$ *design matrix with* $\text{rank}(\mathbf{X}) = p$*, then for every* $y \in \mathbb{R}^n$*,* $\mathbf{H}y$ *(Definition 0.2) is the orthogonal projection of* $y$ *onto* $\mathcal{C}(\mathbf{X})$*. So* $\mathbf{H} = \mathbf{Q}\mathbf{Q}^\top$ *for any* $n \times \tilde{p}$ *matrix* $\mathbf{Q}$ *whose columns are an orthonormal basis of* $\mathcal{C}(\mathbf{X})$*.*

Proof

*Proof.* $\mathbf{H}$ is an orthogonal projection matrix (Theorem 0.2), so $\mathbf{H}y$ is the orthogonal projection of $y$ onto $\mathcal{C}(\mathbf{H})$ (Theorem 0.5, part 1). It remains to show $\mathcal{C}(\mathbf{H}) = \mathcal{C}(\mathbf{X})$.
$\mathcal{C}(\mathbf{H}) \subseteq \mathcal{C}(\mathbf{X})$**.** $\mathbf{H}\underset{\sim}{z} = \mathbf{X}\left((\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\underset{\sim}{z}\right)$ (Theorem 5 in Matrices), which is in $\mathcal{C}(\mathbf{X})$ (Definition 1 in Column Space, Null Space and Rank-Nullity).
$\mathcal{C}(\mathbf{X}) \subseteq \mathcal{C}(\mathbf{H})$**.** First,

$$\begin{aligned}
\mathbf{H}\mathbf{X} &= \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{X} && \text{(Definition 0.2)} \\
&= \mathbf{X}\left((\mathbf{X}^\top\mathbf{X})^{-1}(\mathbf{X}^\top\mathbf{X})\right) && \text{(Theorem 5 in Matrices)} \\
&= \mathbf{X}\,\mathbf{I}_p && \text{(Definition 5 in Special Matrices and Decompositions)} \\
&= \mathbf{X}. && \text{(Theorem 7 in Matrices)}
\end{aligned}$$

So for every $\underset{\sim}{b} \in \mathbb{R}^p$, $\mathbf{X}\underset{\sim}{b} = (\mathbf{H}\mathbf{X})\,\underset{\sim}{b}$, since $\mathbf{H}\mathbf{X} = \mathbf{X}$, and $(\mathbf{H}\mathbf{X})\,\underset{\sim}{b} = \mathbf{H}\,(\mathbf{X}\underset{\sim}{b})$ (Theorem 5 in Matrices), which is in $\mathcal{C}(\mathbf{H})$.
$\mathcal{C}(\mathbf{X})$ has dimension $p \geq 1$ (Theorem 7 in Column Space, Null Space and Rank-Nullity), so an orthonormal basis of it has $p$ vectors (Definition 9 in Subspaces and Rank), and $\mathbf{Q}\mathbf{Q}^\top$ also gives the orthogonal projection onto $\mathcal{C}(\mathbf{X})$ (Theorem 0.4, with $n$ in place of $p$ and $r = p$). By Theorem 0.5, part 2, $\mathbf{H} = \mathbf{Q}\mathbf{Q}^\top$. □

**Example 0.9** (The intercept-only hat matrix, from an orthonormal basis)**.** For the intercept-only design $\mathbf{X} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ of Example 0.2, $\mathcal{C}(\mathbf{X}) = \text{span}\{(1,1)\}$, with orthonormal basis $\underset{\sim}{q}_1 = \frac{1}{\sqrt{2}}(1,1)$, since

$$\begin{aligned}
\|\underset{\sim}{q}_1\| &= \sqrt{\tfrac{1}{2} + \tfrac{1}{2}} \\
&= 1.
\end{aligned}$$

With $\mathbf{Q} = [\underset{\sim}{q}_1]$,

$$\begin{aligned}
\mathbf{Q}\mathbf{Q}^\top &= \tfrac{1}{2}\begin{bmatrix} 1 \\ 1 \end{bmatrix}\begin{bmatrix} 1 & 1 \end{bmatrix} \\
&= \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix},
\end{aligned}$$

the hat matrix found in Example 0.2. The fitted values $(\bar{y}, \bar{y})$ are the closest point to $y$ on the line of constant vectors (Theorem 0.3), and the residuals $(y_1 - \bar{y}, y_2 - \bar{y})$ are orthogonal to $(1, 1)$:

$$\begin{aligned}(1,1) \cdot (y_1 - \bar{y}, y_2 - \bar{y}) &= y_1 + y_2 - 2\bar{y} \\ &= 0,\end{aligned}$$

since $2\bar{y} = y_1 + y_2$.

### Generalized inverses and linear systems

**Source**

This section is adapted from Zhou [5], used under the MIT License (see the license text in Section 2 in Subspaces and Rank). The source defers the existence of the Moore-Penrose inverse to the singular value decomposition; the proof here builds it from a rank factorization (Theorem 15 in Column Space, Null Space and Rank-Nullity) instead, and the section adds the projection $\mathbf{X}\mathbf{G}\mathbf{X}^\top$ for any generalized inverse $\mathbf{G}$ of $\mathbf{X}^\top\mathbf{X}$, which the source does not cover.

**Theorem 0.7** (A scalar factor passes through a matrix product)**.** *For an $m \times k$ matrix $\mathbf{A}$, a $k \times n$ matrix $\mathbf{B}$ and a number $c$,*

$$\begin{aligned}\mathbf{A}\,(c\,\mathbf{B}) &= c\,(\mathbf{A}\mathbf{B}) \\ &= (c\,\mathbf{A})\,\mathbf{B}.\end{aligned}$$

*In particular, $\mathbf{A}\,(\mathbf{B} - \mathbf{C}) = \mathbf{A}\mathbf{B} - \mathbf{A}\mathbf{C}$ and $(\mathbf{B} - \mathbf{C})\,\mathbf{D} = \mathbf{B}\mathbf{D} - \mathbf{C}\mathbf{D}$ for matrices of compatible sizes, where $\mathbf{B} - \mathbf{C} = \mathbf{B} + (-1)\,\mathbf{C}$ (Theorem 4 in Matrices, Definition 6 in Matrices).*

**Proof**

*Proof.* $\mathbf{A}\,(c\,\mathbf{B}) = c\,(\mathbf{A}\mathbf{B})$**.** For each entry $(i, j)$,

$$
\begin{aligned}
(\mathbf{A}\,(c\,\mathbf{B}))_{ij} &= \sum_{l=1}^{k} a_{il}\,(c\,\mathbf{B})_{lj} && \text{(Definition 7 in Matrices)}\\
&= \sum_{l=1}^{k} a_{il}\,(c\,b_{lj}) && \text{(Definition 6 in Matrices)}\\
&= \sum_{l=1}^{k} c\,(a_{il}\,b_{lj}) && \text{(reorder the product of numbers)}\\
&= c\sum_{l=1}^{k} a_{il}\,b_{lj} && \text{(factor } c \text{ out of the sum)}\\
&= c\,(\mathbf{AB})_{ij} && \text{(Definition 7 in Matrices)}\\
&= (c\,(\mathbf{AB}))_{ij}. && \text{(Definition 6 in Matrices)}
\end{aligned}
$$

$(c\,\mathbf{A})\,\mathbf{B} = c\,(\mathbf{AB})$**.** For each entry $(i, j)$,

$$
\begin{aligned}
((c\,\mathbf{A})\,\mathbf{B})_{ij} &= \sum_{l=1}^{k} (c\,a_{il})\,b_{lj} && \text{(Definition 7 in Matrices, Definition 6 in Matrices)}\\
&= c\sum_{l=1}^{k} a_{il}\,b_{lj} && \text{(factor } c \text{ out of the sum)}\\
&= (c\,(\mathbf{AB}))_{ij}. && \text{(Definition 7 in Matrices, Definition 6 in Matrices)}
\end{aligned}
$$

**Differences.**

$$
\begin{aligned}
\mathbf{A}\,(\mathbf{B}-\mathbf{C}) &= \mathbf{A}\,(\mathbf{B}+(-1)\,\mathbf{C}) && \text{(the difference as a sum)}\\
&= \mathbf{AB}+\mathbf{A}\,((-1)\,\mathbf{C}) && \text{(Theorem 6 in Matrices)}\\
&= \mathbf{AB}+(-1)\,(\mathbf{AC}) && \text{(first part)}\\
&= \mathbf{AB}-\mathbf{AC}, && \text{(the difference as a sum)}
\end{aligned}
$$

and $(\mathbf{B}-\mathbf{C})\,\mathbf{D} = \mathbf{BD}-\mathbf{CD}$ the same way, using the second law in Theorem 6 in Matrices and the second part. A column vector is a matrix with one column, so all of this applies with $\mathbf{B}$, $\mathbf{C}$ or $\mathbf{D}$ a vector. □

**Example 0.10** (Pulling a scalar out of a product)**.** With $\mathbf{A} = \begin{bmatrix}1 & 2\end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix}3\\4\end{bmatrix}$:

$$
\begin{aligned}
\mathbf{A}\,(2\,\mathbf{B}) &= \begin{bmatrix}1 & 2\end{bmatrix}\begin{bmatrix}6\\8\end{bmatrix}\\
&= 22,
\end{aligned}
$$

and

$$\begin{aligned} 2\left(\mathbf{AB}\right) &= 2\left(3+8\right) \\ &= 22. \end{aligned}$$

**Definition 0.4** (Generalized inverse)**.** A **generalized inverse** [6] of an $m \times n$ matrix $\mathbf{A}$ is an $n \times m$ matrix $\mathbf{G}$ such that

$$\underset{m\times n}{\mathbf{A}}\ \underset{n\times m}{\mathbf{G}}\ \underset{m\times n}{\mathbf{A}} = \mathbf{A}.$$

**Example 0.11** (A matrix with many generalized inverses)**.** Let $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ and $\mathbf{G} = \begin{bmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{bmatrix}$. Then

$$\begin{aligned} \mathbf{AGA} &= \begin{bmatrix} g_{11} & g_{12} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} && \text{(Definition 7 in Matrices, for } \mathbf{AG}\text{)} \\ &= \begin{bmatrix} g_{11} & 0 \\ 0 & 0 \end{bmatrix}, && \text{(Definition 7 in Matrices)} \end{aligned}$$

which equals $\mathbf{A}$ exactly when $g_{11} = 1$. So every matrix $\begin{bmatrix} 1 & g_{12} \\ g_{21} & g_{22} \end{bmatrix}$ is a generalized inverse of $\mathbf{A}$: a generalized inverse need not be unique. The zero matrix is not one, since its $(1,1)$ entry is 0.

**Theorem 0.8** (An invertible matrix has only one generalized inverse)**.** *If $\mathbf{A}$ is an invertible $p \times p$ matrix (Definition 6 in Special Matrices and Decompositions), its only generalized inverse is $\mathbf{A}^{-1}$.*

Proof

*Proof.*

$$\begin{aligned} \mathbf{A}\mathbf{A}^{-1}\mathbf{A} &= \mathbf{I}_p\mathbf{A} \\ &= \mathbf{A} \end{aligned}$$

(Definition 5 in Special Matrices and Decompositions, Theorem 7 in Matrices), so $\mathbf{A}^{-1}$ is a generalized inverse. If $\mathbf{G}$ is any generalized inverse, then

$$
\begin{aligned}
\mathbf{G} &= \mathbf{I}_p\,\mathbf{G}\,\mathbf{I}_p && \text{(Theorem 7 in Matrices)}\\
&= (\mathbf{A}^{-1}\mathbf{A})\,\mathbf{G}\,(\mathbf{A}\mathbf{A}^{-1}) && \text{(Definition 5 in Special Matrices and Decompositions)}\\
&= \mathbf{A}^{-1}\,(\mathbf{A}\mathbf{G}\mathbf{A})\,\mathbf{A}^{-1} && \text{(Theorem 5 in Matrices)}\\
&= \mathbf{A}^{-1}\mathbf{A}\mathbf{A}^{-1} && \text{(Definition 0.4)}\\
&= \mathbf{I}_p\,\mathbf{A}^{-1} && \text{(Definition 5 in Special Matrices and Decompositions)}\\
&= \mathbf{A}^{-1}. && \text{(Theorem 7 in Matrices)}
\end{aligned}
$$

□

**Example 0.12** (The generalized inverse of an invertible matrix)**.** For $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$ of Example 5 in Special Matrices and Decompositions, the only generalized inverse is $\mathbf{A}^{-1} = \begin{bmatrix} 0.5 & -0.5 \\ 0 & 1 \end{bmatrix}$. The matrix of Example 0.11, by contrast, has infinitely many, so by Theorem 0.8 it is singular.

**Theorem 0.9** (A generalized inverse solves every solvable system)**.** *Let* $\mathbf{A}$ *be* $m \times n$*, let* $\mathbf{G}$ *be a generalized inverse of* $\mathbf{A}$ *(Definition 0.4), and let* $\underset{\sim}{b} \in \mathbb{R}^m$*. If* $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{b}$ *has a solution, then* $\mathbf{G}\underset{\sim}{b}$ *is a solution.*

Proof

*Proof.* Let $\underset{\sim}{x}_0$ be a solution, so $\mathbf{A}\underset{\sim}{x}_0 = \underset{\sim}{b}$. Then

$$
\begin{aligned}
\mathbf{A}\,(\mathbf{G}\underset{\sim}{b}) &= \mathbf{A}\mathbf{G}\,(\mathbf{A}\underset{\sim}{x}_0) && (\text{substitute } \underset{\sim}{b} = \mathbf{A}\underset{\sim}{x}_0)\\
&= (\mathbf{A}\mathbf{G}\mathbf{A})\,\underset{\sim}{x}_0 && \text{(Theorem 5 in Matrices)}\\
&= \mathbf{A}\underset{\sim}{x}_0 && \text{(Definition 0.4)}\\
&= \underset{\sim}{b}. && (\underset{\sim}{x}_0 \text{ is a solution})
\end{aligned}
$$

□

**Example 0.13** (Solving with a generalized inverse)**.** With $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ and its generalized inverse $\mathbf{G} = \begin{bmatrix} 1 & 5 \\ 7 & 0 \end{bmatrix}$ (Example 0.11), the system $\mathbf{A}\underset{\sim}{x} = (2, 0)$ has the solution $(2, 0)$, and $\mathbf{G}\,(2, 0) = (2, 14)$ is a solution too: $\mathbf{A}\,(2, 14) = (2, 0)$. The system $\mathbf{A}\underset{\sim}{x} = (2, 1)$ has no solution, since the second entry of $\mathbf{A}\underset{\sim}{x}$ is always 0; there $\mathbf{G}\,(2, 1) = (7, 14)$ gives $\mathbf{A}\,(7, 14) = (7, 0) \neq (2, 1)$.

**Definition 0.5** (Moore-Penrose inverse)**.** A **Moore-Penrose inverse** [7] of an $m \times n$ matrix $\mathbf{A}$ is an $n \times m$ matrix $\mathbf{G}$ satisfying all four conditions

1. $\mathbf{AGA} = \mathbf{A}$;
2. $\mathbf{GAG} = \mathbf{G}$;
3. $(\mathbf{AG})^\top = \mathbf{AG}$;
4. $(\mathbf{GA})^\top = \mathbf{GA}$.

**Example 0.14** (Checking the four conditions)**.** For $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, the matrix $\mathbf{G} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ satisfies all four:

$$\begin{aligned} \mathbf{AG} &= \mathbf{GA} \\ &= \mathbf{A} \end{aligned}$$

and $\mathbf{A}^2 = \mathbf{A}$, so conditions 1 and 2 read $\mathbf{A} = \mathbf{A}$, and conditions 3 and 4 hold because $\mathbf{A}$ is symmetric.

The generalized inverse $\mathbf{G}' = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$ of $\mathbf{A}$ (Example 0.11) is not a Moore-Penrose inverse: $\mathbf{AG}' = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$, which is not symmetric, so condition 3 fails.

**Theorem 0.10** (A Moore-Penrose inverse is a generalized inverse)**.** *If* $\mathbf{G}$ *is a Moore-Penrose inverse of an* $m \times n$ *matrix* $\mathbf{A}$ *(Definition 0.5), then* $\mathbf{G}$ *is a generalized inverse of* $\mathbf{A}$ *(Definition 0.4).*

Proof

*Proof.* A generalized inverse of $\mathbf{A}$ is an $n \times m$ matrix $\mathbf{G}$ with $\mathbf{AGA} = \mathbf{A}$. A Moore-Penrose inverse is an $n \times m$ matrix, and condition 1 of Definition 0.5 is $\mathbf{AGA} = \mathbf{A}$. □

**Example 0.15** (The Moore-Penrose inverse of a rank-one matrix)**.** The $3 \times 2$ matrix $\mathbf{A}$ below has rank 1, because its second column is twice its first, so it has no ordinary inverse. The function `MASS::ginv()` computes its Moore-Penrose inverse $\mathbf{G}$. The code checks the four conditions of Definition 0.5, of which the first is the equation that makes $\mathbf{G}$ a generalized inverse.

```r
A <- cbind(c(1, 2, 3), c(2, 4, 6))
G <- MASS::ginv(A)
conditions <- c(
  "1: AGA = A" = isTRUE(all.equal(A %*% G %*% A, A)),
  "2: GAG = G" = isTRUE(all.equal(G %*% A %*% G, G)),
  "3: AG symmetric" = isTRUE(all.equal(A %*% G, t(A %*% G))),
  "4: GA symmetric" = isTRUE(all.equal(G %*% A, t(G %*% A)))
)
conditions
#>      1: AGA = A      2: GAG = G 3: AG symmetric 4: GA symmetric
#>             TRUE            TRUE            TRUE            TRUE
round(G, 4)
#>        [,1]   [,2]   [,3]
#> [1,] 0.0143 0.0286 0.0429
#> [2,] 0.0286 0.0571 0.0857
```

All 4 of 4 conditions hold, so $\mathbf{G}$ is a Moore-Penrose inverse of $\mathbf{A}$. By Theorem 0.10 it is also a generalized inverse, which is condition 1: $\mathbf{AGA} = \mathbf{A}$ up to rounding error.

**Theorem 0.11** (A matrix has at most one Moore-Penrose inverse)**.** *If $\mathbf{G}_1$ and $\mathbf{G}_2$ are both Moore-Penrose inverses of an $m \times n$ matrix $\mathbf{A}$ (Definition 0.5), then $\mathbf{G}_1 = \mathbf{G}_2$.*

Proof

*Proof.* $\mathbf{AG}_1 = \mathbf{AG}_2$**.**

$$
\begin{aligned}
\mathbf{AG}_1 &= (\mathbf{AG}_1)^\top && \text{(condition 3 for } \mathbf{G}_1\text{)} \\
&= \mathbf{G}_1{}^\top \mathbf{A}^\top && \text{(Theorem 16 in Matrices)} \\
&= \mathbf{G}_1{}^\top (\mathbf{AG}_2\mathbf{A})^\top && \text{(condition 1 for } \mathbf{G}_2\text{)} \\
&= \mathbf{G}_1{}^\top \mathbf{A}^\top \mathbf{G}_2{}^\top \mathbf{A}^\top && \text{(Theorem 16 in Matrices, twice)} \\
&= (\mathbf{G}_1{}^\top \mathbf{A}^\top)(\mathbf{G}_2{}^\top \mathbf{A}^\top) && \text{(Theorem 5 in Matrices)} \\
&= (\mathbf{AG}_1)^\top (\mathbf{AG}_2)^\top && \text{(Theorem 16 in Matrices, twice)} \\
&= (\mathbf{AG}_1)(\mathbf{AG}_2) && \text{(condition 3 for } \mathbf{G}_1 \text{ and } \mathbf{G}_2\text{)} \\
&= (\mathbf{AG}_1\mathbf{A})\,\mathbf{G}_2 && \text{(Theorem 5 in Matrices)} \\
&= \mathbf{AG}_2. && \text{(condition 1 for } \mathbf{G}_1\text{)}
\end{aligned}
$$

$\mathbf{G}_1\mathbf{A} = \mathbf{G}_2\mathbf{A}$**.**

$$
\begin{aligned}
\mathbf{G}_1\mathbf{A} &= (\mathbf{G}_1\mathbf{A})^\top && \text{(condition 4 for } \mathbf{G}_1\text{)} \\
&= \mathbf{A}^\top \mathbf{G}_1{}^\top && \text{(Theorem 16 in Matrices)} \\
&= (\mathbf{A}\mathbf{G}_2\mathbf{A})^\top \mathbf{G}_1{}^\top && \text{(condition 1 for } \mathbf{G}_2\text{)} \\
&= \mathbf{A}^\top \mathbf{G}_2{}^\top \mathbf{A}^\top \mathbf{G}_1{}^\top && \text{(Theorem 16 in Matrices, twice)} \\
&= (\mathbf{A}^\top \mathbf{G}_2{}^\top)(\mathbf{A}^\top \mathbf{G}_1{}^\top) && \text{(Theorem 5 in Matrices)} \\
&= (\mathbf{G}_2\mathbf{A})^\top (\mathbf{G}_1\mathbf{A})^\top && \text{(Theorem 16 in Matrices, twice)} \\
&= (\mathbf{G}_2\mathbf{A})(\mathbf{G}_1\mathbf{A}) && \text{(condition 4 for } \mathbf{G}_2 \text{ and } \mathbf{G}_1\text{)} \\
&= \mathbf{G}_2(\mathbf{A}\mathbf{G}_1\mathbf{A}) && \text{(Theorem 5 in Matrices)} \\
&= \mathbf{G}_2\mathbf{A}. && \text{(condition 1 for } \mathbf{G}_1\text{)}
\end{aligned}
$$

**Conclusion.**

$$
\begin{aligned}
\mathbf{G}_1 &= \mathbf{G}_1\mathbf{A}\mathbf{G}_1 && \text{(condition 2 for } \mathbf{G}_1\text{)} \\
&= \mathbf{G}_1(\mathbf{A}\mathbf{G}_1) && \text{(Theorem 5 in Matrices)} \\
&= \mathbf{G}_1(\mathbf{A}\mathbf{G}_2) && \text{(the first step)} \\
&= (\mathbf{G}_1\mathbf{A})\mathbf{G}_2 && \text{(Theorem 5 in Matrices)} \\
&= (\mathbf{G}_2\mathbf{A})\mathbf{G}_2 && \text{(the second step)} \\
&= \mathbf{G}_2\mathbf{A}\mathbf{G}_2 && \text{(Theorem 5 in Matrices)} \\
&= \mathbf{G}_2. && \text{(condition 2 for } \mathbf{G}_2\text{)}
\end{aligned}
$$

□

**Example 0.16** (Only one of the many generalized inverses qualifies)**.** The matrix $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ has infinitely many generalized inverses $\begin{bmatrix} 1 & g_{12} \\ g_{21} & g_{22} \end{bmatrix}$ (Example 0.11), and $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ is a Moore-Penrose inverse (Example 0.14). By Theorem 0.11 it is the only one, so every other choice of $g_{12}, g_{21}, g_{22}$ breaks condition 2, 3 or 4. For instance, $g_{21} = 1$ (the rest 0) gives $\mathbf{G}\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$, which is not symmetric.

**Theorem 0.12** (Every matrix has a Moore-Penrose inverse)**.** *Every $m \times n$ matrix $\mathbf{A}$ has a Moore-Penrose inverse (Definition 0.5), written $\mathbf{A}^+$; it is unique by Theorem 0.11.*

- *If $\mathbf{A} = \mathbf{0}_{m\times n}$, then $\mathbf{A}^+ = \mathbf{0}_{n\times m}$.*
- *If $\operatorname{rank}(\mathbf{A}) = r \geq 1$ and $\mathbf{A} = \mathbf{C}\mathbf{R}$ is a rank factorization (Definition 11 in Column Space, Null Space and Rank-Nullity), then*

$$\underbrace{\mathbf{A}^{+}}_{n\times m} = \underbrace{\mathbf{R}^{\top}}_{n\times r}\underbrace{(\mathbf{R}\mathbf{R}^{\top})^{-1}}_{r\times r}\underbrace{(\mathbf{C}^{\top}\mathbf{C})^{-1}}_{r\times r}\underbrace{\mathbf{C}^{\top}}_{r\times m}.$$

Proof

*Proof.* **The two cases cover every matrix.** If $\mathbf{A} \neq \mathbf{0}_{m\times n}$, it has a nonzero column, which is linearly independent on its own ($c\,\underset{\sim}{v} = \underset{\sim}{0}$ with $\underset{\sim}{v} \neq \underset{\sim}{0}$ forces $c = 0$), so $\operatorname{rank}(\mathbf{A}) \geq 1$ (Definition 2 in Subspaces and Rank).
**The zero matrix.** With $\mathbf{A} = \mathbf{0}_{m\times n}$ and $\mathbf{G} = \mathbf{0}_{n\times m}$, every product in conditions 1 to 4 is a zero matrix, and a zero matrix equals its own transpose, so all four conditions hold.
**The two inverses exist.** Now suppose $\operatorname{rank}(\mathbf{A}) = r \geq 1$. A rank factorization exists (Theorem 15 in Column Space, Null Space and Rank-Nullity). $\mathbf{C}$ is $m \times r$ and $\mathbf{R}$ is $r \times n$. Both have rank $r$:

$$\begin{aligned} r &= \operatorname{rank}(\mathbf{C}\mathbf{R}) && (\mathbf{A} = \mathbf{C}\mathbf{R}) \\ &\leq \operatorname{rank}(\mathbf{C}) && \text{(Theorem 9 in Column Space, Null Space and Rank-Nullity)} \\ &\leq r, && \text{(Corollary 1 in Column Space, Null Space and Rank-Nullity, } \mathbf{C} \text{ has } r \text{ columns)} \end{aligned}$$

and, using $(\mathbf{C}\mathbf{R})^{\top} = \mathbf{R}^{\top}\mathbf{C}^{\top}$ (Theorem 16 in Matrices),

$$\begin{aligned} r &= \operatorname{rank}(\mathbf{R}^{\top}\mathbf{C}^{\top}) && \text{(Theorem 10 in Column Space, Null Space and Rank-Nullity)} \\ &\leq \operatorname{rank}(\mathbf{R}^{\top}) && \text{(Theorem 9 in Column Space, Null Space and Rank-Nullity)} \\ &\leq r. && \text{(Corollary 1 in Column Space, Null Space and Rank-Nullity, } \mathbf{R}^{\top} \text{ has } r \text{ columns)} \end{aligned}$$

So $\mathbf{C}$ and $\mathbf{R}^{\top}$ have full column rank $r$, and $\mathbf{C}^{\top}\mathbf{C}$ and $\mathbf{R}\mathbf{R}^{\top} = (\mathbf{R}^{\top})^{\top}\mathbf{R}^{\top}$ are invertible (Theorem 0.1). They are symmetric:

$$\begin{aligned} (\mathbf{C}^{\top}\mathbf{C})^{\top} &= \mathbf{C}^{\top}(\mathbf{C}^{\top})^{\top} \\ &= \mathbf{C}^{\top}\mathbf{C} \end{aligned}$$

(Theorem 16 in Matrices, Definition 3 in Matrices), and likewise for $\mathbf{R}\mathbf{R}^{\top}$. Write $\mathbf{S} \overset{\text{def}}{=} (\mathbf{R}\mathbf{R}^{\top})^{-1}$ and $\mathbf{T} \overset{\text{def}}{=} (\mathbf{C}^{\top}\mathbf{C})^{-1}$, both symmetric (Corollary 1 in Special Matrices and Decompositions), and $\mathbf{G} \overset{\text{def}}{=} \mathbf{R}^{\top}\mathbf{S}\mathbf{T}\mathbf{C}^{\top}$.
**Two products.**

$$\begin{aligned} \mathbf{A}\mathbf{G} &= \mathbf{C}\mathbf{R}\,\mathbf{R}^{\top}\mathbf{S}\mathbf{T}\mathbf{C}^{\top} && \text{(substitute } \mathbf{A} \text{ and } \mathbf{G}) \\ &= \mathbf{C}\,(\mathbf{R}\mathbf{R}^{\top})\,\mathbf{S}\,\mathbf{T}\mathbf{C}^{\top} && \text{(Theorem 5 in Matrices)} \\ &= \mathbf{C}\,\mathbf{I}_r\,\mathbf{T}\mathbf{C}^{\top} && \text{(Definition 5 in Special Matrices and Decompositions)} \\ &= \mathbf{C}\mathbf{T}\mathbf{C}^{\top}, && \text{(Theorem 7 in Matrices)} \end{aligned}$$

$$
\begin{aligned}
\mathbf{GA} &= \mathbf{R}^\top \mathbf{S}\mathbf{T}\mathbf{C}^\top \mathbf{C}\mathbf{R} && \text{(substitute } \mathbf{G} \text{ and } \mathbf{A}\text{)} \\
&= \mathbf{R}^\top \mathbf{S}\,\mathbf{T}\,(\mathbf{C}^\top \mathbf{C})\,\mathbf{R} && \text{(Theorem 5 in Matrices)} \\
&= \mathbf{R}^\top \mathbf{S}\,\mathbf{I}_r\,\mathbf{R} && \text{(Definition 5 in Special Matrices and Decompositions)} \\
&= \mathbf{R}^\top \mathbf{S}\mathbf{R}. && \text{(Theorem 7 in Matrices)}
\end{aligned}
$$

**Conditions 3 and 4.**

$$
\begin{aligned}
(\mathbf{AG})^\top &= (\mathbf{C}\mathbf{T}\mathbf{C}^\top)^\top && \text{(first product)} \\
&= (\mathbf{C}\,(\mathbf{T}\mathbf{C}^\top))^\top && \text{(Theorem 5 in Matrices)} \\
&= (\mathbf{T}\mathbf{C}^\top)^\top \mathbf{C}^\top && \text{(Theorem 16 in Matrices)} \\
&= (\mathbf{C}^\top)^\top \mathbf{T}^\top \mathbf{C}^\top && \text{(Theorem 16 in Matrices)} \\
&= \mathbf{C}\,\mathbf{T}^\top \mathbf{C}^\top && \text{(Definition 3 in Matrices)} \\
&= \mathbf{C}\mathbf{T}\mathbf{C}^\top, && (\mathbf{T} \text{ is symmetric}) \\
(\mathbf{GA})^\top &= (\mathbf{R}^\top \mathbf{S}\mathbf{R})^\top && \text{(second product)} \\
&= (\mathbf{R}^\top\,(\mathbf{S}\mathbf{R}))^\top && \text{(Theorem 5 in Matrices)} \\
&= (\mathbf{S}\mathbf{R})^\top\,(\mathbf{R}^\top)^\top && \text{(Theorem 16 in Matrices)} \\
&= \mathbf{R}^\top\,\mathbf{S}^\top\,(\mathbf{R}^\top)^\top && \text{(Theorem 16 in Matrices)} \\
&= \mathbf{R}^\top\,\mathbf{S}^\top\,\mathbf{R} && \text{(Definition 3 in Matrices)} \\
&= \mathbf{R}^\top \mathbf{S}\mathbf{R}. && (\mathbf{S} \text{ is symmetric})
\end{aligned}
$$

**Condition 1.**

$$
\begin{aligned}
\mathbf{AGA} &= \mathbf{C}\mathbf{T}\mathbf{C}^\top \mathbf{C}\mathbf{R} && \text{(first product)} \\
&= \mathbf{C}\,\mathbf{T}\,(\mathbf{C}^\top \mathbf{C})\,\mathbf{R} && \text{(Theorem 5 in Matrices)} \\
&= \mathbf{C}\,\mathbf{I}_r\,\mathbf{R} && \text{(Definition 5 in Special Matrices and Decompositions)} \\
&= \mathbf{A}. && \text{(Theorem 7 in Matrices)}
\end{aligned}
$$

**Condition 2.**

$$
\begin{aligned}
\mathbf{GAG} &= \mathbf{R}^\top \mathbf{S}\mathbf{R}\,\mathbf{R}^\top \mathbf{S}\mathbf{T}\mathbf{C}^\top && \text{(second product)} \\
&= \mathbf{R}^\top\,\mathbf{S}\,(\mathbf{R}\mathbf{R}^\top)\,\mathbf{S}\,\mathbf{T}\mathbf{C}^\top && \text{(Theorem 5 in Matrices)} \\
&= \mathbf{R}^\top\,\mathbf{I}_r\,\mathbf{S}\,\mathbf{T}\mathbf{C}^\top && \text{(Definition 5 in Special Matrices and Decompositions)} \\
&= \mathbf{G}. && \text{(Theorem 7 in Matrices)}
\end{aligned}
$$

□

**Example 0.17** (The Moore-Penrose inverse of a rank-one matrix)**.** For $\mathbf{A}$ in Example 1 in Column Space, Null Space and Rank-Nullity, the rank factorization $\mathbf{C} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$, $\mathbf{R} = \begin{bmatrix} 1 & -2 & -2 \end{bmatrix}$ (Example 24 in Column Space, Null Space and Rank-Nullity) gives

$$\begin{aligned} \mathbf{C}^\top \mathbf{C} &= 1 + 9 \\ &= 10 \end{aligned}$$

and

$$\begin{aligned} \mathbf{R}\mathbf{R}^\top &= 1 + 4 + 4 \\ &= 9, \end{aligned}$$

so

$$\begin{aligned} \mathbf{A}^+ &= \frac{1}{9 \cdot 10} \begin{bmatrix} 1 \\ -2 \\ -2 \end{bmatrix} \begin{bmatrix} 1 & 3 \end{bmatrix} \\ &= \frac{1}{90} \begin{bmatrix} 1 & 3 \\ -2 & -6 \\ -2 & -6 \end{bmatrix}. \end{aligned}$$

As a check on condition 1,

$$\begin{aligned} \mathbf{A}\mathbf{A}^+ &= \mathbf{C}\mathbf{T}\mathbf{C}^\top \\ &= \frac{1}{10} \begin{bmatrix} 1 & 3 \\ 3 & 9 \end{bmatrix}, \end{aligned}$$

and its first row times $\mathbf{A}$ is $\frac{1}{10}(1 \cdot (1, -2, -2) + 3 \cdot (3, -6, -6)) = (1, -2, -2)$, the first row of $\mathbf{A}$; the second row is 3 times the first, like the second row of $\mathbf{A}$.

**Definition 0.6** (Consistent linear system)**.** A linear system (Definition 14 in Matrices) $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{b}$, with $\mathbf{A}$ an $m \times n$ matrix and $\underset{\sim}{b} \in \mathbb{R}^m$, is **consistent** [8] if it has at least one solution $\underset{\sim}{x} \in \mathbb{R}^n$, and **inconsistent** otherwise.

**Example 0.18** (A consistent system and an inconsistent one)**.** With $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, $\mathbf{A}(x_1, x_2) = (x_1, 0)$ (Definition 13 in Matrices).

- $\mathbf{A}\underset{\sim}{x} = (2, 0)$ is consistent: $\underset{\sim}{x} = (2, 0)$ is a solution.
- $\mathbf{A}\underset{\sim}{x} = (2, 1)$ is inconsistent: the second entry of $\mathbf{A}\underset{\sim}{x}$ is always 0, never 1.

**Theorem 0.13** (When a linear system has a solution)**.** *Let* $\mathbf{A}$ *be* $m \times n$*, let* $\underset{\sim}{b} \in \mathbb{R}^m$*, and let* $\mathbf{G}$ *be any generalized inverse of* $\mathbf{A}$ *(Definition 0.4). The following statements are equivalent:*

1. $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{b}$ *is consistent (Definition 0.6);*
2. $\underset{\sim}{b} \in \mathcal{C}(\mathbf{A})$ *(Definition 1 in Column Space, Null Space and Rank-Nullity);*
3. $\mathbf{AG}\underset{\sim}{b} = \underset{\sim}{b}$.

Proof

*Proof.* **1 and 2 are equivalent.** Statement 1 says $\underset{\sim}{b} = \mathbf{A}\underset{\sim}{x}$ for some $\underset{\sim}{x}$ (Definition 0.6), and $\mathcal{C}(\mathbf{A})$ is the set of vectors $\mathbf{A}\underset{\sim}{x}$ (Definition 1 in Column Space, Null Space and Rank-Nullity), so $\underset{\sim}{b}$ is in it exactly when $\underset{\sim}{b} = \mathbf{A}\underset{\sim}{x}$ for some $x$.
**1 implies 3.** By Theorem 0.9, $\mathbf{G}\underset{\sim}{b}$ is a solution, so $\mathbf{A}\left(\mathbf{G}\underset{\sim}{b}\right) = \underset{\sim}{b}$, which is statement 3 (Theorem 5 in Matrices).
**3 implies 1.** If $\mathbf{AG}\underset{\sim}{b} = \underset{\sim}{b}$, then $\underset{\sim}{x} = \mathbf{G}\underset{\sim}{b}$ is a solution, since $\mathbf{A}\left(\mathbf{G}\underset{\sim}{b}\right) = (\mathbf{AG})\,\underset{\sim}{b}$ (Theorem 5 in Matrices). □

**Example 0.19** (Testing consistency with a generalized inverse)**.** For $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ and the generalized inverse $\mathbf{G} = \begin{bmatrix} 1 & 5 \\ 7 & 0 \end{bmatrix}$, $\mathbf{AG} = \begin{bmatrix} 1 & 5 \\ 0 & 0 \end{bmatrix}$.

- $\underset{\sim}{b} = (2, 0)$:

$$\begin{aligned} \mathbf{AG}\underset{\sim}{b} &= (2, 0) \\ &= \underset{\sim}{b}, \end{aligned}$$

  so the system is consistent.

- $\underset{\sim}{b} = (2, 1)$: $\mathbf{AG}\underset{\sim}{b} = (7, 0) \neq \underset{\sim}{b}$, so it is not, as Example 0.13 found directly.

**Theorem 0.14** (All solutions of a linear system)**.** *Let* $\mathbf{A}$ *be* $m \times n$ *with a generalized inverse* $\mathbf{G}$ *(Definition 0.4).*

1. $\mathcal{N}(\mathbf{A}) = \mathcal{C}(\mathbf{I}_n - \mathbf{GA})$*:* $\underset{\sim}{x}$ *solves* $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{0}_m$ *exactly when* $\underset{\sim}{x} = (\mathbf{I}_n - \mathbf{GA})\,\underset{\sim}{q}$ *for some* $q \in \mathbb{R}^n$.
2. *If* $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{b}$ *is consistent (Definition 0.6), then* $\underset{\sim}{x}$ *is a solution exactly when*

$$\underset{\sim}{x} = \mathbf{G}\underset{\sim}{b} + (\mathbf{I}_n - \mathbf{GA})\,\underset{\sim}{q} \quad \text{for some } \underset{\sim}{q} \in \mathbb{R}^n :$$

   *one particular solution plus a vector of* $\mathcal{N}(\mathbf{A})$.

Proof

*Proof.* **Part 1.** First,

$$\begin{aligned}
\mathbf{A}(\mathbf{I}_n - \mathbf{GA}) &= \mathbf{AI}_n - \mathbf{A}(\mathbf{GA}) && \text{(Theorem 0.7)}\\
&= \mathbf{AI}_n - \mathbf{AGA} && \text{(Theorem 5 in Matrices)}\\
&= \mathbf{A} - \mathbf{AGA} && \text{(Theorem 7 in Matrices)}\\
&= \mathbf{A} - \mathbf{A} && \text{(Definition 0.4)}\\
&= \mathbf{0}_{m\times n}, && \text{(arithmetic)}
\end{aligned}$$

so $\mathbf{A}(\mathbf{I}_n - \mathbf{GA})\,\underset{\sim}{q} = \underset{\sim}{0}_m$ for every $\underset{\sim}{q}$ (Theorem 5 in Matrices), and $\mathcal{C}(\mathbf{I}_n - \mathbf{GA}) \subseteq \mathcal{N}(\mathbf{A})$. Conversely, if $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{0}_m$, then

$$\begin{aligned}
(\mathbf{I}_n - \mathbf{GA})\,\underset{\sim}{x} &= \mathbf{I}_n\underset{\sim}{x} - (\mathbf{GA})\,\underset{\sim}{x} && \text{(Theorem 0.7, with } \underset{\sim}{x} \text{ as the last factor)}\\
&= \underset{\sim}{x} - (\mathbf{GA})\,\underset{\sim}{x} && \text{(Theorem 7 in Matrices)}\\
&= \underset{\sim}{x} - \mathbf{G}(\mathbf{A}\underset{\sim}{x}) && \text{(Theorem 5 in Matrices)}\\
&= \underset{\sim}{x} - \mathbf{G}\,\underset{\sim}{0}_m && (\mathbf{A}\underset{\sim}{x} = \underset{\sim}{0}_m)\\
&= \underset{\sim}{x}, && (\mathbf{G}\,\underset{\sim}{0}_m = \underset{\sim}{0}_n)
\end{aligned}$$

so $\underset{\sim}{x}$ is in $\mathcal{C}(\mathbf{I}_n - \mathbf{GA})$, with $\underset{\sim}{q} = \underset{\sim}{x}$.
**Part 2.** $\mathbf{G}\underset{\sim}{b}$ is a solution (Theorem 0.9). For any $\underset{\sim}{x}$,

$$\begin{aligned}
\mathbf{A}(\underset{\sim}{x} - \mathbf{G}\underset{\sim}{b}) &= \mathbf{A}\underset{\sim}{x} - \mathbf{A}(\mathbf{G}\underset{\sim}{b}) && \text{(Theorem 4 in Subspaces and Rank, with coefficients 1 and } -1)\\
&= \mathbf{A}\underset{\sim}{x} - \mathbf{AG}\underset{\sim}{b} && \text{(Theorem 5 in Matrices)}\\
&= \mathbf{A}\underset{\sim}{x} - \underset{\sim}{b}, && \text{(Theorem 0.13, statement 3)}
\end{aligned}$$

so $\underset{\sim}{x}$ solves $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{b}$ exactly when $\underset{\sim}{x} - \mathbf{G}\underset{\sim}{b} \in \mathcal{N}(\mathbf{A})$, that is, by part 1, exactly when $\underset{\sim}{x} - \mathbf{G}\underset{\sim}{b} = (\mathbf{I}_n - \mathbf{GA})\,\underset{\sim}{q}$ for some $\underset{\sim}{q}$. □

**Example 0.20** (All solutions of a consistent system)**.** For $\mathbf{A} = \begin{bmatrix}1 & 0\\ 0 & 0\end{bmatrix}$, $\mathbf{G} = \begin{bmatrix}1 & 5\\ 7 & 0\end{bmatrix}$ and $\underset{\sim}{b} = (2,0)$: $\mathbf{G}\underset{\sim}{b} = (2,14)$, $\mathbf{GA} = \begin{bmatrix}1 & 0\\ 7 & 0\end{bmatrix}$, and $\mathbf{I}_2 - \mathbf{GA} = \begin{bmatrix}0 & 0\\ -7 & 1\end{bmatrix}$, so

$$\underset{\sim}{x} = (2,14) + (0, -7q_1 + q_2), \qquad q_1, q_2 \in \mathbb{R}.$$

Since $-7q_1 + q_2$ can be any number, the solutions are exactly the vectors $(2,t)$, $t \in \mathbb{R}$, which is what

$$\begin{aligned}
\mathbf{A}(x_1, x_2) &= (x_1, 0)\\
&= (2,0)
\end{aligned}$$

says directly.

**Corollary 0.1** (Solvable for every right-hand side, and solvable uniquely)**.** *Let* $\mathbf{A}$ *be* $m \times n$*.*

1. $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{b}$ *is consistent for every* $\underset{\sim}{b} \in \mathbb{R}^m$ *exactly when* $\text{rank}(\mathbf{A}) = m$*.*
2. *A consistent system* $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{b}$ *has exactly one solution exactly when* $\text{rank}(\mathbf{A}) = n$*.*

Proof

*Proof.* **Part 1.** By Theorem 0.13, the system is consistent for every $\underset{\sim}{b}$ exactly when $\mathcal{C}(\mathbf{A}) = \mathbb{R}^m$. $\mathcal{C}(\mathbf{A})$ is a subspace of $\mathbb{R}^m$ (Theorem 1 in Column Space, Null Space and Rank-Nullity) with dimension $\text{rank}(\mathbf{A})$ (Theorem 7 in Column Space, Null Space and Rank-Nullity), and $\dim(\mathbb{R}^m) = m$ (Example 16 in Subspaces and Rank). If the dimension is $m$, then $\mathcal{C}(\mathbf{A}) = \mathbb{R}^m$ (Theorem 11 in Subspaces and Rank); if $\mathcal{C}(\mathbf{A}) = \mathbb{R}^m$, the dimension is $m$.
**Part 2.** By Theorem 0.14, the solutions are a particular solution plus the vectors of $\mathcal{N}(\mathbf{A})$, so there is exactly one solution exactly when $\mathcal{N}(\mathbf{A}) = \{\underset{\sim}{0}_n\}$, that is, when $\text{nullity}(\mathbf{A}) = 0$ (a subspace of dimension 0 has the empty list as a basis, whose span is $\{\underset{\sim}{0}\}$, by Definition 9 in Subspaces and Rank and Definition 5 in Subspaces and Rank; and $\{\underset{\sim}{0}\}$ has dimension 0, by Example 16 in Subspaces and Rank). By Theorem 8 in Column Space, Null Space and Rank-Nullity, that is when $\text{rank}(\mathbf{A}) = n$. □

**Example 0.21** (Three systems)**.**

- The $2 \times 3$ matrix $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$ of Example 16 in Column Space, Null Space and Rank-Nullity has rank $2 = m$, so every system with it is consistent, but $2 < 3 = n$, so the solutions are never unique.
- The $3 \times 2$ matrix $\mathbf{X} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$ of Example 0.1 has rank $2 = n$, so a consistent system has one solution, but $2 < 3 = m$, so some systems are inconsistent: $(1, 0, 0) \notin \mathcal{C}(\mathbf{X})$, because $a\,(1, 1, 1) + b\,(1, 2, 3)$ has equal differences between consecutive entries, and $(1, 0, 0)$ does not.
- An invertible $p \times p$ matrix $\mathbf{A}$ has rank $p$: if $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{0}$ then

$$\begin{aligned} \underset{\sim}{x} &= \mathbf{A}^{-1}\mathbf{A}\underset{\sim}{x} \\ &= \underset{\sim}{0}, \end{aligned}$$

so its nullity is 0 and its rank is $p$ (Theorem 8 in Column Space, Null Space and Rank-Nullity). So every system with it is consistent (part 1, with $p = m$) and has exactly one solution (part 2, with $p = n$), namely $\mathbf{A}^{-1}\underset{\sim}{b}$, since $\mathbf{A}\left(\mathbf{A}^{-1}\underset{\sim}{b}\right) = \underset{\sim}{b}$.

**Theorem 0.15** (The projection onto a column space, with any generalized inverse)**.** *Let* $\mathbf{X}$ *be an* $n \times p$ *matrix of any rank, and let* $\mathbf{G}$ *be any generalized inverse of* $\mathbf{X}^\top\mathbf{X}$ *(Definition 0.4). Then for every* $y \in \mathbb{R}^n$, $\mathbf{X}\mathbf{G}\mathbf{X}^\top y$ *is the orthogonal projection of* $y$ *onto* $\mathcal{C}(\mathbf{X})$ *(Definition 0.3). So the matrix* $\mathbf{X}\mathbf{G}\mathbf{X}^\top$ *is the same for every choice of* $\mathbf{G}$*, and when* $\text{rank}(\mathbf{X}) = p$ *it is the hat matrix* $\mathbf{H}$ *(Definition 0.2).*

Proof

*Proof.* **$\mathbf{X}^\top y$ is in $\mathcal{C}(\mathbf{X}^\top\mathbf{X})$.** $\mathcal{C}(\mathbf{X}^\top\mathbf{X}) \subseteq \mathcal{C}(\mathbf{X}^\top)$, since $(\mathbf{X}^\top\mathbf{X})\underset{\sim}{z} = \mathbf{X}^\top(\mathbf{X}\underset{\sim}{z})$ (Theorem 5 in Matrices). Both are subspaces of $\mathbb{R}^p$ (Theorem 1 in Column Space, Null Space and Rank-Nullity) with the same dimension: $\text{rank}(\mathbf{X}^\top\mathbf{X}) = \text{rank}(\mathbf{X}^\top)$ (Theorem 10 in Column Space, Null Space and Rank-Nullity, Theorem 7 in Column Space, Null Space and Rank-Nullity). So $\mathcal{C}(\mathbf{X}^\top\mathbf{X}) = \mathcal{C}(\mathbf{X}^\top)$ (Theorem 11 in Subspaces and Rank), and since $\mathbf{X}^\top y \in \mathcal{C}(\mathbf{X}^\top)$ (Definition 1 in Column Space, Null Space and Rank-Nullity), $\mathbf{X}^\top\underset{\sim}{y} = \mathbf{X}^\top\mathbf{X}\underset{\sim}{\tilde{z}}$ for some $\underset{\sim}{\tilde{z}} \in \mathbb{R}^p$.

**The split.** Let

$$\begin{aligned}\underset{\sim}{u} &= \mathbf{X}\mathbf{G}\mathbf{X}^\top y \\ &= \mathbf{X}\left(\mathbf{G}\mathbf{X}^\top\underset{\sim}{\tilde{y}}\right)\end{aligned}$$

(Theorem 5 in Matrices), which is in $\mathcal{C}(\mathbf{X})$ (Definition 1 in Column Space, Null Space and Rank-Nullity). For $\underset{\sim}{y} - \underset{\sim}{u}$,

$$\begin{aligned}
\mathbf{X}^\top(\underset{\sim}{y} - \underset{\sim}{u}) &= \mathbf{X}^\top\underset{\sim}{y} - \mathbf{X}^\top\underset{\sim}{u} && \text{(Theorem 4 in Subspaces and Rank, with coefficients 1 and } -1\text{)} \\
&= \mathbf{X}^\top\underset{\sim}{\tilde{y}} - \mathbf{X}^\top(\mathbf{X}\mathbf{G}\mathbf{X}^\top\underset{\sim}{\tilde{y}}) && \text{(substitute } \underset{\sim}{u}\text{)} \\
&= \mathbf{X}^\top\underset{\sim}{\tilde{y}} - (\mathbf{X}^\top\mathbf{X})\,\mathbf{G}\,(\mathbf{X}^\top\underset{\sim}{\tilde{y}}) && \text{(Theorem 5 in Matrices)} \\
&= \mathbf{X}^\top\underset{\sim}{\tilde{y}} - (\mathbf{X}^\top\mathbf{X})\,\mathbf{G}\,(\mathbf{X}^\top\mathbf{X})\,\underset{\sim}{\tilde{z}} && \text{(first step)} \\
&= \mathbf{X}^\top\underset{\sim}{\tilde{y}} - ((\mathbf{X}^\top\mathbf{X})\,\mathbf{G}\,(\mathbf{X}^\top\mathbf{X}))\,\underset{\sim}{\tilde{z}} && \text{(Theorem 5 in Matrices)} \\
&= \mathbf{X}^\top\underset{\sim}{\tilde{y}} - (\mathbf{X}^\top\mathbf{X})\,\underset{\sim}{\tilde{z}} && \text{(Definition 0.4)} \\
&= \mathbf{X}^\top\underset{\sim}{\tilde{y}} - \mathbf{X}^\top\underset{\sim}{y} && \text{(first step)} \\
&= \underset{\sim}{0}_p, && \text{(arithmetic)}
\end{aligned}$$

so $\underset{\sim}{y} - \underset{\sim}{u} \in \mathcal{N}(\mathbf{X}^\top) = \mathcal{C}(\mathbf{X})^\perp$ (Theorem 7 in Direct Sums and Orthogonal Complements). By the uniqueness in Theorem 8 in Direct Sums and Orthogonal Complements, $\underset{\sim}{u}$ is the orthogonal projection of $\underset{\sim}{y}$ onto $\mathcal{C}(\mathbf{X})$.

**Same matrix.** Every choice of $\mathbf{G}$ gives a matrix that projects every $y$ onto $\mathcal{C}(\mathbf{X})$, so all choices give the same matrix (Theorem 0.5, part 2). When $\text{rank}(\mathbf{X}) = p$, $(\mathbf{X}^\top\mathbf{X})^{-1}$ is one choice (Theorem 0.1, Theorem 0.8), and it gives $\mathbf{H}$. □

**Example 0.22** (A design matrix with a repeated column)**.** Let $\mathbf{X} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, an intercept column entered twice, so $\text{rank}(\mathbf{X}) = 1 < 2$ and $\mathbf{X}^\top\mathbf{X} = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$ is not invertible: its rank is $\text{rank}(\mathbf{X}) = 1 < 2$ (Theorem 10 in Column Space, Null Space and Rank-Nullity), while an invertible $2 \times 2$ matrix has rank 2 (Example 0.21). The matrix $\mathbf{G} = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & 0 \end{bmatrix}$ is a generalized inverse of it:

$$
\begin{aligned}
(\mathbf{X}^\top\mathbf{X})\,\mathbf{G}\,(\mathbf{X}^\top\mathbf{X}) &= \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix} && \text{(Definition 7 in Matrices, for the first product)} \\
&= \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}. && \text{(Definition 7 in Matrices)}
\end{aligned}
$$

Then

$$
\begin{aligned}
\mathbf{X}\mathbf{G}\mathbf{X}^\top &= \begin{bmatrix} \frac{1}{2} & 0 \\ \frac{1}{2} & 0 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} && \text{(Definition 7 in Matrices, for } \mathbf{X}\mathbf{G}\text{)} \\
&= \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}, && \text{(Definition 7 in Matrices)}
\end{aligned}
$$

the hat matrix of the intercept-only model (Example 0.2). Another generalized inverse, $\mathbf{G}' = \begin{bmatrix} 0 & 0 \\ 0 & \frac{1}{2} \end{bmatrix}$, which is one because $(\mathbf{X}^\top\mathbf{X})\,\mathbf{G}' = \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}$ and that times $\mathbf{X}^\top\mathbf{X}$ is again $\begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$, gives $\mathbf{X}\mathbf{G}' = \begin{bmatrix} 0 & \frac{1}{2} \\ 0 & \frac{1}{2} \end{bmatrix}$ and the same $\mathbf{X}\mathbf{G}'\mathbf{X}^\top = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}$, as Theorem 0.15 says it must. Both project onto $\mathcal{C}(\mathbf{X}) = \text{span}\{(1, 1)\}$, the column space of the intercept-only design.

### Solving linear systems and least squares

> **Source**
>
> This section is adapted from Zhou [5] and Zhou [9], used under the MIT License (see the license text in Section 2 in Subspaces and Rank). The first source states that every invertible matrix has an LU factorization; that is false without reordering rows (the $2 \times 2$ row swap $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ has none), so this section states only how an LU factorization, when one exists, solves a system. The least squares results are proved here from the closest-point theorem (Theorem 0.3) rather than by differentiating, and the QR factorization is derived from Gram-Schmidt (Theorem 10 in Inner Products and Orthogonality).

**Definition 0.7** (Upper triangular matrix)**.** A square matrix $\mathbf{U}$ is **upper triangular** [10] if every entry below the diagonal is 0: $u_{ij} = 0$ whenever $i > j$.

**Definition 0.8** (Lower triangular matrix)**.** A square matrix $\mathbf{L}$ is **lower triangular** if every entry above the diagonal is 0: $\ell_{ij} = 0$ whenever $i < j$.

**Definition 0.9** (Unit lower triangular matrix)**.** A lower triangular matrix (Definition 0.8) whose diagonal entries are all 1 is **unit lower triangular**.

**Example 0.23** (Triangular and not)**.**

- $\begin{bmatrix} 2 & 1 & -1 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & -1 \end{bmatrix}$ is upper triangular: its entries below the diagonal, in positions $(2,1)$, $(3,1)$ and $(3,2)$, are 0.
- $\begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix}$ is unit lower triangular.
- A diagonal matrix (Definition 4 in Special Matrices and Decompositions) is both upper and lower triangular.
- $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ is neither: its $(2,1)$ entry is not 0, and neither is its $(1,2)$ entry.

**Theorem 0.16** (Solving a triangular system by substitution)**.** *Let $\mathbf{U}$ be an $n \times n$ upper triangular matrix (Definition 0.7) whose diagonal entries $u_{11}, \dots, u_{nn}$ are all nonzero. For every $\underset{\sim}{b} \in \mathbb{R}^n$, the system $\mathbf{U}\underset{\sim}{x} = \underset{\sim}{b}$ has exactly one solution, given for $i = n, n-1, \dots, 1$ in*

*turn by*

$$x_i = \frac{1}{u_{ii}}\left(b_i - \sum_{j=i+1}^{n} u_{ij}\, x_j\right)$$

*(**back substitution**). Likewise, if* $\mathbf{L}$ *is lower triangular with nonzero diagonal,* $\mathbf{L}\underset{\sim}{y} = \underset{\sim}{b}$ *has exactly one solution, given for* $i = 1, \dots, n$ *by* $y_i = \frac{1}{\ell_{ii}}\left(b_i - \sum_{j=1}^{i-1} \ell_{ij}\, y_j\right)$ *(**forward substitution**).*

Proof

*Proof.* Row $i$ of $\mathbf{U}\underset{\sim}{x} = \underset{\sim}{b}$ reads

$$\begin{aligned} b_i &= \sum_{j=1}^{n} u_{ij}\, x_j && \text{(Definition 13 in Matrices)} \\ &= \sum_{j=i}^{n} u_{ij}\, x_j && (u_{ij} = 0 \text{ for } j < i) \\ &= u_{ii}\, x_i + \sum_{j=i+1}^{n} u_{ij}\, x_j. && \text{(split off the } j = i \text{ term)} \end{aligned}$$

This equation is equivalent to each of

$$\begin{aligned} u_{ii}\, x_i &= b_i - \sum_{j=i+1}^{n} u_{ij}\, x_j && \text{(subtract the sum from both sides)} \\ x_i &= \frac{1}{u_{ii}}\left(b_i - \sum_{j=i+1}^{n} u_{ij}\, x_j\right). && \text{(divide by } u_{ii} \neq 0) \end{aligned}$$

Row $n$ involves only $x_n$, so it fixes $x_n$; once $x_{i+1}, \dots, x_n$ are fixed, row $i$ fixes $x_i$. So the rows, taken from the last to the first, hold exactly when $\underset{\sim}{x}$ is the vector the formula builds: there is one solution, and only one. For $\mathbf{L}$, row $i$ reads $\sum_{j=1}^{i-1} \ell_{ij}\, y_j + \ell_{ii}\, y_i = b_i$, and the same argument runs from the first row to the last. □

**Example 0.24** (Back substitution on a $3 \times 3$ system)**.** Solve $\begin{bmatrix} 2 & 1 & -1 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & -1 \end{bmatrix} \underset{\sim}{x} = \begin{bmatrix} 8 \\ 1 \\ 1 \end{bmatrix}$:

1. $$\begin{aligned} x_3 &= \frac{1}{-1} \cdot 1 \\ &= -1; \end{aligned}$$

2.
$$\begin{aligned} x_2 &= \frac{1}{1/2}\left(1 - \tfrac{1}{2}\cdot(-1)\right) \\ &= 2 \cdot \tfrac{3}{2} \\ &= 3; \end{aligned}$$

3.
$$\begin{aligned} x_1 &= \frac{1}{2}(8 - 1\cdot 3 - (-1)(-1)) \\ &= \frac{1}{2}\cdot 4 \\ &= 2. \end{aligned}$$

So $\underset{\sim}{x} = (2, 3, -1)$. Without a nonzero diagonal the method fails: $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \underset{\sim}{x} = (1, 1)$ has no solution, since its second row reads $0 = 1$.

**Theorem 0.17** (QR factorization)**.** *Let* $\mathbf{A}$ *be an* $m \times n$ *matrix with* $\operatorname{rank}(\mathbf{A}) = n$*. Then*

$$\underset{m\times n}{\mathbf{A}} = \underset{m\times n}{\mathbf{Q}}\ \underset{n\times n}{\mathbf{R}},$$

*where the columns of* $\mathbf{Q}$ *are orthonormal (Definition 19 in Vectors) and* $\mathbf{R}$ *is upper triangular (Definition 0.7) with positive diagonal entries.*

Proof

*Proof.* The $n$ columns $\underset{\sim}{a}_1, \ldots, \underset{\sim}{a}_n$ of $\mathbf{A}$ are linearly independent (Definition 3 in Subspaces and Rank), so the Gram-Schmidt process (Definition 11 in Inner Products and Orthogonality) runs all $n$ steps and gives orthonormal $\underset{\sim}{q}_1, \ldots, \underset{\sim}{q}_n$ (Theorem 10 in Inner Products and Orthogonality, parts 1 and 3); let $\mathbf{Q}$ have these columns. At step $i$, $\underset{\sim}{q}'_i \neq \underset{\sim}{0}$, and rearranging the orthogonalize step,

$$\begin{aligned} \underset{\sim}{a}_i &= \underset{\sim}{q}'_i + \sum_{j=1}^{i-1} (\underset{\sim}{q}_j \cdot \underset{\sim}{a}_i)\, \underset{\sim}{q}_j && \text{(add the sum to both sides of the orthogonalize step)} \\ &= \|\underset{\sim}{q}'_i\|\, \underset{\sim}{q}_i + \sum_{j=1}^{i-1} (\underset{\sim}{q}_j \cdot \underset{\sim}{a}_i)\, \underset{\sim}{q}_j. && \text{(normalize step: } \underset{\sim}{q}'_i = \|\underset{\sim}{q}'_i\|\, \underset{\sim}{q}_i) \end{aligned}$$

Let $\mathbf{R}$ be the $n \times n$ matrix with $r_{ji} = \underset{\sim}{q}_j \cdot \underset{\sim}{a}_i$ for $j < i$, $r_{ii} = \|\underset{\sim}{q}'_i\| > 0$, and $r_{ji} = 0$ for $j > i$; it is upper triangular with positive diagonal. Column $i$ of $\mathbf{QR}$ is $\mathbf{Q}$ times column $i$ of $\mathbf{R}$ (Definition 7 in Matrices), and

$$
\begin{aligned}
\mathbf{Q}\,(r_{1i},\dots,r_{ni}) &= \sum_{j=1}^{n} r_{ji}\,\underset{\sim}{q}_j && \text{(Theorem 3 in Subspaces and Rank)}\\
&= \sum_{j=1}^{i} r_{ji}\,\underset{\sim}{q}_j && (r_{ji}=0 \text{ for } j>i)\\
&= r_{ii}\,\underset{\sim}{q}_i + \sum_{j=1}^{i-1} r_{ji}\,\underset{\sim}{q}_j && \text{(split off the } j=i \text{ term)}\\
&= \|\underset{\sim}{q}'_i\|\,\underset{\sim}{q}_i + \sum_{j=1}^{i-1} (\underset{\sim}{q}_j\cdot\underset{\sim}{a}_i)\,\underset{\sim}{q}_j && \text{(the entries of } \mathbf{R})\\
&= \underset{\sim}{a}_i. && \text{(the display)}
\end{aligned}
$$

So $\mathbf{QR}=\mathbf{A}$. □

**Example 0.25** (A QR factorization from Gram-Schmidt)**.** For $\mathbf{A}$ with columns $\underset{\sim}{a}_1=(1,1,0)$, $\underset{\sim}{a}_2=(1,0,1)$, $\underset{\sim}{a}_3=(0,1,1)$, Example 21 in Inner Products and Orthogonality and Example 23 in Inner Products and Orthogonality found $\underset{\sim}{q}_1=\frac{1}{\sqrt{2}}(1,1,0)$, $\underset{\sim}{q}_2=\frac{1}{\sqrt{6}}(1,-1,2)$, $\underset{\sim}{q}_3=\frac{1}{\sqrt{3}}(-1,1,1)$, with $\|\underset{\sim}{q}'_1\|=\sqrt{2}$, $\|\underset{\sim}{q}'_2\|=\sqrt{3/2}$, $\|\underset{\sim}{q}'_3\|=2/\sqrt{3}$,

$$
\begin{aligned}
\underset{\sim}{q}_1\cdot\underset{\sim}{a}_2 &= \underset{\sim}{q}_1\cdot\underset{\sim}{a}_3\\
&= \tfrac{1}{\sqrt{2}}
\end{aligned}
$$

and $\underset{\sim}{q}_2\cdot\underset{\sim}{a}_3=\frac{1}{\sqrt{6}}$. So

$$
\mathbf{R}=\begin{bmatrix} \sqrt{2} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & \sqrt{3/2} & \frac{1}{\sqrt{6}} \\ 0 & 0 & \frac{2}{\sqrt{3}} \end{bmatrix}.
$$

As a check on the second column,

$$
\begin{aligned}
\tfrac{1}{\sqrt{2}}\,\underset{\sim}{q}_1+\sqrt{3/2}\,\underset{\sim}{q}_2 &= \tfrac{1}{2}(1,1,0)+\tfrac{1}{2}(1,-1,2)\\
&= (1,0,1)\\
&= \underset{\sim}{a}_2.
\end{aligned}
$$

**Definition 0.10** (Least squares solution)**.** Let $\mathbf{A}$ be $m\times n$ and $\underset{\sim}{b}\in\mathbb{R}^m$. A **least squares solution** [11] of $\mathbf{A}\underset{\sim}{x}=\underset{\sim}{b}$ is a vector $\hat{\underset{\sim}{x}}\in\mathbb{R}^n$ such that

$$
\|\underset{\sim}{b}-\mathbf{A}\hat{\underset{\sim}{x}}\|\le\|\underset{\sim}{b}-\mathbf{A}\underset{\sim}{x}\| \quad \text{for every } \underset{\sim}{x}\in\mathbb{R}^n.
$$

**Example 0.26** (Fitting a constant)**.** Let $\mathbf{A} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\underset{\sim}{b} = (1, 3)$. The system $x\,(1, 1) = (1, 3)$ is inconsistent (Definition 0.6): its first entry needs $x = 1$ and its second needs $x = 3$. For any number $x$,

$$\begin{aligned} \|\underset{\sim}{b} - \mathbf{A}x\|^2 &= (1 - x)^2 + (3 - x)^2 \\ &= 2\,(x - 2)^2 + 2, \end{aligned}$$

which is smallest at $x = 2$, so $\hat{x} = 2$ is the least squares solution. The value $x = 1$ is not one: it gives $0 + 4 = 4 > 2$. The normal equations read $2x = 4$, and their solution is $x = 2$ too.

**Definition 0.11** (Normal equations)**.** For $\mathbf{A}$ and $\underset{\sim}{b}$ as in Definition 0.10, the equations $\mathbf{A}^\top \mathbf{A}\underset{\sim}{x} = \mathbf{A}^\top \underset{\sim}{b}$ are the **normal equations** [12].

**Theorem 0.18** (Least squares solutions solve the normal equations)**.** *Let* $\mathbf{A}$ *be* $m \times n$ *and* $\underset{\sim}{b} \in \mathbb{R}^m$*.*

1. $\underset{\sim}{\hat{x}}$ *is a least squares solution (Definition 0.10) exactly when it solves the normal equations* $\mathbf{A}^\top \mathbf{A}\underset{\sim}{\hat{x}} = \mathbf{A}^\top \underset{\sim}{b}$*.*
2. *The normal equations are consistent (Definition 0.6).*
3. *Every least squares solution gives the same fitted vector* $\mathbf{A}\underset{\sim}{\hat{x}}$*: the orthogonal projection of* $\underset{\sim}{b}$ *onto* $\mathcal{C}(\mathbf{A})$ *(Definition 0.3).*
4. *The least squares solution is unique exactly when* $\operatorname{rank}(\mathbf{A}) = n$*, and then* $\underset{\sim}{\hat{x}} = (\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top \underset{\sim}{b}$*.*

Proof

*Proof.* Let $\underset{\sim}{u}$ be the orthogonal projection of $\underset{\sim}{b}$ onto $\mathcal{C}(\mathbf{A})$. The vectors $\mathbf{A}\underset{\sim}{x}$, as $\underset{\sim}{x}$ ranges over $\mathbb{R}^n$, are exactly the points of $\mathcal{C}(\mathbf{A})$ (Definition 1 in Column Space, Null Space and Rank-Nullity). **Least squares solutions are the solutions of** $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{u}$**.** If $\mathbf{A}\underset{\sim}{\hat{x}} = \underset{\sim}{u}$, then for every $\underset{\sim}{x}$, $\mathbf{A}\underset{\sim}{x} \in \mathcal{C}(\mathbf{A})$, so $\|\underset{\sim}{b} - \mathbf{A}\underset{\sim}{\hat{x}}\| = \|\underset{\sim}{b} - \underset{\sim}{u}\| \le \|\underset{\sim}{b} - \mathbf{A}\underset{\sim}{x}\|$ (Theorem 0.3), and $\underset{\sim}{\hat{x}}$ is a least squares solution. Conversely, let $\underset{\sim}{\hat{x}}$ be a least squares solution. $\underset{\sim}{u} \in \mathcal{C}(\mathbf{A})$, so $\underset{\sim}{u} = \mathbf{A}\underset{\sim}{x_0}$ for some $\underset{\sim}{x_0}$ (Definition 1 in Column Space, Null Space and Rank-Nullity), and

$$\begin{aligned} \|\underset{\sim}{b} - \mathbf{A}\underset{\sim}{\hat{x}}\| &\le \|\underset{\sim}{b} - \mathbf{A}\underset{\sim}{x_0}\| && \text{(Definition 0.10)} \\ &= \|\underset{\sim}{b} - \underset{\sim}{u}\| && (\underset{\sim}{u} = \mathbf{A}\underset{\sim}{x_0}) \\ &\le \|\underset{\sim}{b} - \mathbf{A}\underset{\sim}{\hat{x}}\|, && \text{(Theorem 0.3, with } \underset{\sim}{w} = \mathbf{A}\underset{\sim}{\hat{x}} \in \mathcal{C}(\mathbf{A})) \end{aligned}$$

so the last inequality is an equality, and the equality case of Theorem 0.3 gives $\mathbf{A}\underset{\sim}{\hat{x}} = \underset{\sim}{u}$. **Parts 1 and 3.** $\mathbf{A}\underset{\sim}{\hat{x}} = \underset{\sim}{u}$ holds exactly when $\underset{\sim}{b} - \mathbf{A}\underset{\sim}{\hat{x}} \in \mathcal{C}(\mathbf{A})^\perp$: if $\mathbf{A}\underset{\sim}{\hat{x}} = \underset{\sim}{u}$, then

$\underset{\sim}{b} - \underset{\sim}{u} \in \mathcal{C}(\mathbf{A})^\perp$ (Definition 0.3); conversely, if $\underset{\sim}{b} - \mathbf{A}\underset{\sim}{\hat{x}} \in \mathcal{C}(\mathbf{A})^\perp$, then $\underset{\sim}{b} = \mathbf{A}\underset{\sim}{\hat{x}} + (\underset{\sim}{b} - \mathbf{A}\underset{\sim}{\hat{x}})$ is a split into $\mathcal{C}(\mathbf{A})$ and $\mathcal{C}(\mathbf{A})^\perp$, so $\mathbf{A}\underset{\sim}{\hat{x}} = \underset{\sim}{u}$ by the uniqueness in Theorem 8 in Direct Sums and Orthogonal Complements. And $\mathcal{C}(\mathbf{A})^\perp = \mathcal{N}(\mathbf{A}^\top)$ (Theorem 7 in Direct Sums and Orthogonal Complements), where

$$\begin{aligned}\mathbf{A}^\top(\underset{\sim}{b} - \mathbf{A}\underset{\sim}{\hat{x}}) &= \mathbf{A}^\top\underset{\sim}{b} - \mathbf{A}^\top(\mathbf{A}\underset{\sim}{\hat{x}}) && \text{(Theorem 0.7)}\\ &= \mathbf{A}^\top\underset{\sim}{b} - \mathbf{A}^\top\mathbf{A}\underset{\sim}{\hat{x}}, && \text{(Theorem 5 in Matrices)}\end{aligned}$$

which is $\underset{\sim}{0}_n$ exactly when the normal equations hold.

**Part 2.** The vector $\underset{\sim}{x}_0$ above satisfies $\mathbf{A}\underset{\sim}{x}_0 = \underset{\sim}{u}$, so it is a least squares solution, and it solves the normal equations by part 1.

**Part 4.** The least squares solutions are exactly the solutions of $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{u}$, a consistent system, so there is exactly one when $\operatorname{rank}(\mathbf{A}) = n$ and more than one otherwise (Corollary 0.1). When $\operatorname{rank}(\mathbf{A}) = n$, $\mathbf{A}^\top\mathbf{A}$ is invertible (Theorem 0.1), and with $\mathbf{M} \overset{\text{def}}{=} \mathbf{A}^\top\mathbf{A}$,

$$\begin{aligned}\underset{\sim}{\hat{x}} &= \mathbf{I}_n\,\underset{\sim}{\hat{x}} && \text{(Theorem 7 in Matrices)}\\ &= (\mathbf{M}^{-1}\mathbf{M})\,\underset{\sim}{\hat{x}} && \text{(Definition 5 in Special Matrices and Decompositions)}\\ &= \mathbf{M}^{-1}(\mathbf{M}\underset{\sim}{\hat{x}}) && \text{(Theorem 5 in Matrices)}\\ &= \mathbf{M}^{-1}\mathbf{A}^\top\underset{\sim}{b}. && \text{(the normal equations)}\end{aligned}$$

□

**Example 0.27** (A least squares line)**.** Fit $y = x_1 + x_2 t$ to the points $(t, y) = (1,1), (2,2), (3,2)$: $\mathbf{A} = \begin{bmatrix}1 & 1\\ 1 & 2\\ 1 & 3\end{bmatrix}$ (the $\mathbf{X}$ of Example 0.1) and $\underset{\sim}{b} = (1,2,2)$. $\mathbf{A}$ has rank 2 (Example 2 in Subspaces and Rank), so the solution is unique (part 4). $\mathbf{A}^\top\mathbf{A} = \begin{bmatrix}3 & 6\\ 6 & 14\end{bmatrix}$ has inverse $\frac{1}{6}\begin{bmatrix}14 & -6\\ -6 & 3\end{bmatrix}$ (Example 0.1), and

$$\begin{aligned}\mathbf{A}^\top\underset{\sim}{b} &= (1+2+2,\ 1+4+6)\\ &= (5, 11),\end{aligned}$$

so

$$\begin{aligned}\underset{\sim}{\hat{x}} &= \frac{1}{6}(14\cdot 5 - 6\cdot 11,\ -6\cdot 5 + 3\cdot 11)\\ &= \frac{1}{6}(4, 3)\\ &= (\tfrac{2}{3}, \tfrac{1}{2}).\end{aligned}$$

The fitted values are $\mathbf{A}\underset{\sim}{\hat{x}} = (\frac{7}{6}, \frac{5}{3}, \frac{13}{6})$, and the residuals $\underset{\sim}{b} - \mathbf{A}\underset{\sim}{\hat{x}} = (-\frac{1}{6}, \frac{1}{3}, -\frac{1}{6})$ are orthogonal to both columns of $\mathbf{A}$: $-\frac{1}{6} + \frac{1}{3} - \frac{1}{6} = 0$ and $-\frac{1}{6} + \frac{2}{3} - \frac{1}{2} = 0$.

**Example 0.28** (Many least squares solutions, one fitted vector)**.** With the rank-1 matrix $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 1 & 2 \\ 1 & 2 \end{bmatrix}$ (Example 2 in Subspaces and Rank) and $\underset{\sim}{b} = (1, 2, 2)$, $\mathbf{A}^\top \mathbf{A} = \begin{bmatrix} 3 & 6 \\ 6 & 12 \end{bmatrix}$ and $\mathbf{A}^\top \underset{\sim}{b} = (5, 10)$, so both normal equations say $3x_1 + 6x_2 = 5$, that is, $x_1 + 2x_2 = \frac{5}{3}$. There are infinitely many least squares solutions, such as $(\frac{5}{3}, 0)$ and $(0, \frac{5}{6})$, as part 4 predicts for $\operatorname{rank}(\mathbf{A}) = 1 < 2$; but every one gives the same fitted vector

$$\begin{aligned} \mathbf{A}\underset{\sim}{\hat{x}} &= (x_1 + 2x_2)\,(1, 1, 1) \\ &= \tfrac{5}{3}\,(1, 1, 1), \end{aligned}$$

as part 3 says.

**Theorem 0.19** (Solving least squares by QR)**.** *Let* $\mathbf{A}$ *be* $m \times n$ *with* $\operatorname{rank}(\mathbf{A}) = n$*, with QR factorization* $\mathbf{A} = \mathbf{QR}$ *(Theorem 0.17), and let* $\underset{\sim}{b} \in \mathbb{R}^m$*. The least squares solution (Definition 0.10) is the unique solution of*

$$\underset{n\times n}{\mathbf{R}}\ \underset{\sim}{\hat{x}} = \underset{n\times m}{\mathbf{Q}^\top}\ \underset{\sim}{b},$$

*which back substitution computes (Theorem 0.16). When* $m = n$*, it is the unique solution of* $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{b}$*.*

Proof

*Proof.* $\mathbf{R}$ is upper triangular with positive diagonal, so $\mathbf{R}\underset{\sim}{x} = \mathbf{Q}^\top \underset{\sim}{b}$ has exactly one solution $\underset{\sim}{\hat{x}}$ (Theorem 0.16). $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_n$, as in the proof of Theorem 0.4. Then

$$\begin{aligned} \mathbf{A}^\top \mathbf{A}\underset{\sim}{\hat{x}} &= (\mathbf{QR})^\top (\mathbf{QR})\,\underset{\sim}{\hat{x}} && \text{(substitute } \mathbf{A} = \mathbf{QR}) \\ &= \mathbf{R}^\top \mathbf{Q}^\top (\mathbf{QR})\,\underset{\sim}{\hat{x}} && \text{(Theorem 16 in Matrices)} \\ &= \mathbf{R}^\top (\mathbf{Q}^\top \mathbf{Q})\,(\mathbf{R}\underset{\sim}{\hat{x}}) && \text{(Theorem 5 in Matrices)} \\ &= \mathbf{R}^\top \mathbf{I}_n\,(\mathbf{R}\underset{\sim}{\hat{x}}) && (\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_n) \\ &= \mathbf{R}^\top (\mathbf{R}\underset{\sim}{\hat{x}}) && \text{(Theorem 7 in Matrices)} \\ &= \mathbf{R}^\top (\mathbf{Q}^\top \underset{\sim}{b}) && (\mathbf{R}\underset{\sim}{\hat{x}} = \mathbf{Q}^\top \underset{\sim}{b}) \\ &= (\mathbf{R}^\top \mathbf{Q}^\top)\,\underset{\sim}{b} && \text{(Theorem 5 in Matrices)} \\ &= (\mathbf{QR})^\top \underset{\sim}{b} && \text{(Theorem 16 in Matrices)} \\ &= \mathbf{A}^\top \underset{\sim}{b}, && (\mathbf{A} = \mathbf{QR}) \end{aligned}$$

so $\underset{\sim}{\hat{x}}$ solves the normal equations and is the least squares solution (Theorem 0.18), which is unique because $\operatorname{rank}(\mathbf{A}) = n$. When $m = n$,

$$\begin{aligned}\operatorname{rank}(\mathbf{A}) &= n \\ &= m,\end{aligned}$$

so $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{b}$ is consistent for every $\underset{\sim}{b}$ (Corollary 0.1, part 1), so its solution leaves residual $\underset{\sim}{0}$, is a least squares solution, and is therefore $\underset{\sim}{\hat{x}}$. □

**Example 0.29** (The least squares line by QR)**.** For $\mathbf{A}$ and $\underset{\sim}{b}$ of Example 0.27, Gram-Schmidt gives $\underset{\sim}{q}_1 = \frac{1}{\sqrt{3}}(1,1,1)$; then

$$\begin{aligned}\underset{\sim}{q}_1 \cdot (1,2,3) &= \tfrac{6}{\sqrt{3}} \\ &= 2\sqrt{3},\end{aligned}$$

$$\begin{aligned}\underset{\sim}{q}_2' &= (1,2,3) - 2\,(1,1,1) \\ &= (-1,0,1)\end{aligned}$$

with norm $\sqrt{2}$, and $\underset{\sim}{q}_2 = \frac{1}{\sqrt{2}}(-1,0,1)$. So

$$\begin{aligned}\mathbf{R} &= \begin{bmatrix}\sqrt{3} & 2\sqrt{3} \\ 0 & \sqrt{2}\end{bmatrix}, \\ \mathbf{Q}^\top \underset{\sim}{b} &= \left(\tfrac{1+2+2}{\sqrt{3}},\ \tfrac{-1+0+2}{\sqrt{2}}\right) \\ &= \left(\tfrac{5}{\sqrt{3}}, \tfrac{1}{\sqrt{2}}\right).\end{aligned}$$

Back substitution gives

$$\begin{aligned}x_2 &= \tfrac{1}{\sqrt{2}} \cdot \tfrac{1}{\sqrt{2}} \\ &= \tfrac{1}{2}\end{aligned}$$

and

$$\begin{aligned}x_1 &= \tfrac{1}{\sqrt{3}}\left(\tfrac{5}{\sqrt{3}} - 2\sqrt{3} \cdot \tfrac{1}{2}\right) \\ &= \tfrac{5}{3} - 1 \\ &= \tfrac{2}{3},\end{aligned}$$

the solution found in Example 0.27.

**Definition 0.12** (LU factorization)**.** An **LU factorization** [13] of an $n \times n$ matrix $\mathbf{A}$ is a product $\mathbf{A} = \mathbf{LU}$ with $\mathbf{L}$ unit lower triangular (Definition 0.9) and $\mathbf{U}$ upper triangular (Definition 0.7).

**Example 0.30** (An LU factorization of a $3 \times 3$ matrix)**.** Let $\mathbf{A} = \begin{bmatrix} 2 & 1 & -1 \\ -3 & -1 & 2 \\ -2 & 1 & 2 \end{bmatrix}$, $\mathbf{L} = \begin{bmatrix} 1 & 0 & 0 \\ -\frac{3}{2} & 1 & 0 \\ -1 & 4 & 1 \end{bmatrix}$ (unit lower triangular) and $\mathbf{U} = \begin{bmatrix} 2 & 1 & -1 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & -1 \end{bmatrix}$ (upper triangular). By Definition 7 in Matrices, entry $(i, j)$ of $\mathbf{LU}$ is $\sum_k \ell_{ik}\, u_{kj}$, so row $i$ of $\mathbf{LU}$ is $\sum_k \ell_{ik}$ times row $k$ of $\mathbf{U}$. Row 1 is $1 \cdot (2, 1, -1) = (2, 1, -1)$; row 2 is $-\frac{3}{2}(2, 1, -1) + (0, \frac{1}{2}, \frac{1}{2}) = (-3, -1, 2)$, and row 3 is $-(2, 1, -1) + 4\,(0, \frac{1}{2}, \frac{1}{2}) + (0, 0, -1) = (-2, 1, 2)$. These are the rows of $\mathbf{A}$, so $\mathbf{LU} = \mathbf{A}$ is an LU factorization. The below-diagonal entries of $\mathbf{L}$ are the negatives of the multiples of earlier rows that Gaussian elimination adds to reduce $\mathbf{A}$ to $\mathbf{U}$.

**Example 0.31** (An invertible matrix with no LU factorization)**.** $\mathbf{P} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ is invertible: $\mathbf{P}^2 = \mathbf{I}_2$, so $\mathbf{P}^{-1} = \mathbf{P}$ (Definition 5 in Special Matrices and Decompositions); but it has no LU factorization. If $\mathbf{P} = \mathbf{LU}$, the $(1, 1)$ entry gives $0 = 1 \cdot u_{11}$, so $u_{11} = 0$, and then the $(2, 1)$ entry gives

$$\begin{aligned} 1 &= \ell_{21}\, u_{11} \\ &= 0, \end{aligned}$$

which is impossible. Swapping the two rows first removes the obstacle: the swapped matrix is $\mathbf{I}_2 = \mathbf{I}_2\, \mathbf{I}_2$.

**Theorem 0.20** (Solving a system with an LU factorization)**.** *If $\mathbf{A} = \mathbf{LU}$ is an LU factorization (Definition 0.12) and the diagonal entries of $\mathbf{U}$ are all nonzero, then for every $\underset{\sim}{b} \in \mathbb{R}^n$ the system $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{b}$ has exactly one solution: solve $\mathbf{L}\underset{\sim}{y} = \underset{\sim}{b}$ by forward substitution, then $\mathbf{U}\underset{\sim}{x} = \underset{\sim}{y}$ by back substitution (Theorem 0.16).*

Proof

*Proof.* $\mathbf{L}$ has diagonal entries 1 and $\mathbf{U}$ has nonzero ones, so each of the two triangular systems has exactly one solution (Theorem 0.16). If $\underset{\sim}{y}$ and $\underset{\sim}{x}$ are those solutions, then

$$\begin{aligned} \mathbf{A}\underset{\sim}{x} &= \mathbf{L}\,(\mathbf{U}\underset{\sim}{x}) \\ &= \mathbf{L}\underset{\sim}{y} \\ &= \underset{\sim}{b} \end{aligned}$$

(Theorem 5 in Matrices), so $\underset{\sim}{x}$ is a solution. Conversely, if $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{b}$, then $\underset{\sim}{y} \overset{\text{def}}{=} \mathbf{U}\underset{\sim}{x}$ satisfies

$$\begin{aligned}\mathbf{L}\underset{\sim}{y} &= \mathbf{L}\left(\mathbf{U}\underset{\sim}{x}\right)\\ &= (\mathbf{LU})\,\underset{\sim}{x}\\ &= \mathbf{A}\underset{\sim}{x}\\ &= \underset{\sim}{b}\end{aligned}$$

(Theorem 5 in Matrices), so $\underset{\sim}{y}$ is the unique solution of that system, and $\underset{\sim}{x}$ is then the unique solution of $\mathbf{U}\underset{\sim}{x} = \underset{\sim}{y}$. □

**Example 0.32** (Two triangular solves)**.** With $\mathbf{L}$ and $\mathbf{U}$ of Example 0.30, solve $\mathbf{A}\underset{\sim}{x} = (8, -11, -3)$. Forward substitution on $\mathbf{L}\underset{\sim}{y} = (8, -11, -3)$ gives $y_1 = 8$,

$$\begin{aligned}y_2 &= -11 + \tfrac{3}{2}\cdot 8\\ &= 1,\end{aligned}$$

$$\begin{aligned}y_3 &= -3 + 8 - 4\cdot 1\\ &= 1.\end{aligned}$$

Back substitution on $\mathbf{U}\underset{\sim}{x} = (8, 1, 1)$ is Example 0.24, which gives $\underset{\sim}{x} = (2, 3, -1)$. As a check,

$$\begin{aligned}\mathbf{A}\,(2, 3, -1) &= (4 + 3 + 1,\ -6 - 3 - 2,\ -4 + 3 - 2)\\ &= (8, -11, -3).\end{aligned}$$


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